Local Interpretation and Green Demand Shocks

October 19, 2026

1 Introduction

Scheduled firm announcements account for a small share of the information that moves stock prices (Ball and Shivakumar, 2008). Much of the rest arrives as events that are not objectively and unambiguously connected to any single firm: tariffs, a pandemic, a hurricane reach a stock through what the investor takes the event to mean for it, and the same event is read differently by different investors (Kandel and Pearson, 1995). Whether this reading shapes portfolio choices is hard to study, because across events the content, the assets affected and the speed of the reaction differ as well.

Extreme weather events provide one event read in many places. A flood or hurricane is a dated, located episode experienced at once in many counties, each with its own investors and its own local information environment, and whether it is a manifestation of climate change is contested. After Hurricane Ian, liberal and conservative respondents differed by 21 to 41 percentage points in associating the storm with climate change, and liberal and conservative outlets differed in whether they suggested or negated the link (Djourelova et al., 2024). A single episode also says little about the climate trend on its own (Choi, Gao, and Jiang, 2020), so what it is taken to mean is the reading it receives. Read as a local hazard, the event concerns only the firms exposed to it, and managers near a disaster reduce their holdings of those firms (Alok, Kumar, and Wermers, 2020).[1] Read as corroborating evidence of climate change, it might concern sustainable investments, a category that held $3.9 trillion at the end of 2025 (Morningstar, 2026), spans firms and places far from the damage, and whose demand shifts account for about 86% of the average annual return on a long-short ESG portfolio (van der Beck, 2026). In this paper, I investigate how the interpretation of an event shapes the direction of investors' portfolio responses. I examine whether reading the same disaster as evidence of climate change, holding the disaster and the attention it receives fixed, moves the households and fund managers who experience it toward sustainable assets, and whether that response reflects their beliefs about climate risk or the politics and incentives of the investors.

In the setting of extreme weather events, I show that the same physical event can receive different interpretations across affected places and use this variation to study portfolio reallocation by geographically identified institutional investors, focusing on household-serving advisers and mutual fund managers. I train an ensemble of large language models and use them to classify nearly 1.5 million articles from 527 U.S. news outlets. I combine these classifications with ZIP- and county-level newspaper circulation data, reaching 2,134 counties, to construct a novel measure of national and local interpretation, validated to separately identify simple attention to a disaster from coverage explicitly interpreting the event as evidence or a manifestation

[1] Even then, which firms an event reaches and how fast investors price it are not given by the event: after the 2016 election, investors incorporated the growth and tax consequences for firms quickly and took longer to work out the trade consequences (Wagner, Zeckhauser, and Ziegler, 2018).

of climate change. I study two settings, local investment advisers serving household clients and actively managed mutual funds. Both let me observe portfolio adjustment at the level of an individual investor in a known county, link it to the local coverage of each disaster, and, for funds, to the political giving of the investment team.

Comparing counties affected by hurricanes and floods, I find that for portfolios managed by local advisers serving household clients, a ten-percentage-point higher share of local news coverage explicitly linking the occurring flood to climate change is associated with a 19.30-basis-point larger increase in the green-ETF share over two quarters, about $5.04 million reallocated to green ETFs at the average county ETF portfolio size, or 19.6% of average pre-event green-ETF holdings. Higher disaster coverage and attention alone lead to greater overall turnover, but it's the interpretation of the disaster as being climate-change related that leads to reallocation specifically toward green assets. The relationship extends from advisers' green ETFs to flows into ESG mutual funds: among funds headquartered in affected states, net inflows into ESG funds relative to non-ESG funds are larger after disasters with high climate interpretation, by 8.49 percentage points of pre-disaster assets over two quarters and 18.64 over three, with no such difference in neighboring unaffected states.

The challenge in identifying this response is that a disaster can move green holdings without any reading of it as climate change. Disasters can strike places where climate concern and green investment are already rising. The damage can create liquidity needs, and the attention a disaster receives can move portfolios whatever its coverage says. Advisers' own views and incentives can move the portfolios they manage for households. The design rules out the main of these channels. I compare changes in portfolios before and after a disaster between affected and unaffected counties linked to the same episode, and within the same disaster I compare investors at the same distance from it whose local coverage reads it more or less as climate change, conditional on disaster attention. The response is measured at pre-event prices, net of valuation changes. Pre-disaster changes in the green share do not differ between affected and unaffected counties, and the estimate is unchanged with local trends and non-disaster climate attention. It is not explained by the county's prior climate concern, political orientation, income or education, it does not appear around earthquakes, and green ETF assets rise together with non-green ETF assets while total holdings do not change, so liquidation does not produce it. The local response does not change when I add the national coverage and the national climate interpretation of the same disaster, and it is similar after years of strong and weak green-ETF returns. The response varies with the political composition of the county and not with the adviser's own political orientation.

Across outlet–episode pairs, stable differences between outlets, such as the political views and green attitudes of their audiences and owners, account for 5.5% of the variance of local climate interpretation, the characteristics of the event for 40.7%, and how a given outlet reads a given disaster for 54.1%. The response to interpretation is the same in counties with more and fewer residents worried about global warming, and does not vary with the county's perceived scientific consensus, Democratic vote share or college share.

The interpretation measure already captures the reading of the event that forms locally, whether the county's outlets supply it or cater to their readers' predisposition. To further investigate the role of locally supplied models and narratives (Schwartzstein and Sunderam, 2021), I use supply-side variation in interpretation. Many local newspapers belong to chains, which write an article once and print it across their newspapers, and under a larger owner the climate-interpreted share of local disaster coverage is lower. The instrument is the size of the chain that owned the county's outlets before the event. It varies at sell-offs,

purchases and mergers of those outlets, as well as upstream mergers and acquisitions of their owners.[2] These transactions follow declining advertising revenue, the liquidity needs of legacy owners and cost cutting in roll-ups (Ewens et al., 2022; Gao et al., 2020), and not the climate attitudes of the local newsroom or of the county's readers. Per ten percentage points of the interpretation share, the instrumented response is 22 to 25 basis points for counties and 19 to 22 basis points for advisers, in line with the local-projection estimates.

To further close the gap between the local interpretation an investor is exposed to and the beliefs of the investor who makes the decision, I turn to actively managed mutual funds, where I observe the investment team that makes the allocation decision directly. Political donations provide information on the prior views of the managers and investment teams responsible for portfolio allocation, management locations give the local interpretation each team is exposed to at the office where the portfolio is run, and separating active stock reallocation from the proportional adjustment of holdings to investor flows isolates the managers' own decisions from those of fund investors. Funds without an ESG mandate have no contractual requirement to invest in green assets, so their reallocation is a look into active rebalancing. The hypothesis is that managers learn about climate change from how a nearby disaster is covered where they work. I find that a ten-percentage-point higher share of climate-linked coverage reaching managers within 50 miles of a disaster raises their active tilt toward greener firms by 69.38 basis points in the quarter of the disaster, while the volume of disaster coverage alone does not move the tilt. The response is located at the fund's managing office: nearby managers reallocate whether or not the headquarters is near the event, and funds whose headquarters is near the event but whose managers are distant show no response. For both investor groups, the response weakens with distance from the event. Detailed holdings further allow me to compare allocation changes within the same security, ensuring that the results are not driven by differences in the types of securities held by managers who differ in their climate beliefs or incentives, and absorbing the stock's own reaction to the event, its return and fundamentals, the firm's location and exposure and the shock to its industry, so that what remains across holders of the same stock is how each manager reads the event.

I then provide suggestive evidence on the mechanism, using survey responses and insurance purchases. In survey data, expectations of imminent or ongoing harm from global warming are positively associated with local advisers' green ETF holdings and rise after floods among moderates and conservatives, with weaker evidence for beliefs about human causation and for policy attitudes. Non-flood extreme weather events predict stronger flood-insurance take-up in affected counties only where local newspapers link them to climate change, a cross-hazard response that salience of the experienced hazard alone does not produce. Together, these patterns point to perceived climate harm as the link between disaster experience and green investment, and to local interpretation as the channel through which the experience becomes perceived harm.

At the national level, noteworthy national disasters raise flows into green funds and the relative returns of sustainable assets (Marshall et al., 2021; Fiordelisi et al., 2023; Misev and Balles, 2024), and flows into ESG portfolios are an important driver of their returns (van der Beck, 2026). The local responses above reach prices when many investors share the reading of a disaster at the same time. I provide suggestive evidence that the climate interpretation of national disaster coverage, beyond the attention a disaster receives, is associated with flows into ESG funds. At the average national interpretation share of 25.6%, a one-standard-deviation national disaster-attention surprise is associated with an estimated cumulative increase in flows into ESG-labelled funds relative to other funds equivalent to approximately $288 million over the attention month and the following two months. The estimated three-month response is approximately $580 million larger at the 75th than at the 25th percentile of the national interpretation share. To gauge the economic significance

[2]For example, the 2019 merger of Gannett and GateHouse.

of interpretation, I consider a counterfactual in which disasters throughout a year collectively activate the institutional investor base, with each investor responding once as the local households represented in my RIA sample do. Holding attention fixed, an interquartile increase in climate framing implies 33.92 basis points of one-time ESG-stock appreciation relative to other stocks, using multipliers calibrated from van der Beck's (2026) model and assuming comparable portfolio demand and market accommodation. This is equivalent in magnitude to 18.2% of the average annual return contribution attributed to institutional ESG demand and 15.2% of the observed annual return spread. Evidence of similar effects of interpretation on active stock reallocation by mutual-fund managers motivates extending the exercise from flows into labeled ESG funds to institutional equity demand.

1.1 Related Literature

Political views and financial responses to common events. This paper contributes to research on political views and financial responses to common events, which documents partisan differences in investment beliefs and capital allocation (Meeuwis et al., 2022; Kempf et al., 2023; Cookson, Engelberg, and Mullins, 2020). Here, I focus on studying the effect of local narratives and interpretations beyond political views, while also examining the interplay between the two. My constructed measure captures geographic differences and changes over time in the explanations linking experience to climate change, as well as shared interpretations of specific events that more stable, cross-sectional differences in political views alone cannot capture.

Experience, media, and climate attitudes. The paper is related to a broader literature on how experience, political views, and media shape climate attitudes (Deryugina, 2013; Ash et al., 2023; Djourelova et al., 2024). My contribution is to document how the local interpretation of disaster experience is reflected in consequential financial decisions, specifically through active portfolio reallocations toward green assets. Complementary survey evidence also examines how responses to disaster experience vary with political views and local media coverage, distinguishing perceived climate risks from beliefs about human causation and policy preferences. These patterns suggest that risk perceptions can respond to experience among conservatives despite persistent political differences in climate attitudes.[3] Read alongside the portfolio responses, they provide suggestive evidence on the role of perceived risk in prompting green investment adjustments.

Disasters, climate attention and financial markets. The paper also relates to research linking climate attention to trading and asset returns (Choi, Gao, and Jiang, 2020; Pástor, Stambaugh, and Taylor, 2022; Ardia et al., 2023).[4] I contribute by identifying local portfolio responses linked to disaster experience and examining how these responses depend on whether local narratives attribute the event to climate change. I therefore distinguish the attention an event receives from its attribution to climate change and trace how geographically dispersed investors translate disaster experience and local information into portfolio adjustments. Disaster coverage is associated with greater gross turnover, while climate attribution is associated with reallocation toward green assets. These findings distinguish local trading activity from the direction of portfolio adjustment.

[3] This pattern is consistent with Deryugina's (2013) finding that abnormally warm weather makes conservatives more likely to report that the effects of global warming have already begun.

[4] Choi, Gao, and Jiang (2020) associate unusually warm local temperatures with increased attention to climate change, retail selling of carbon-intensive stocks, and their relative underperformance. Pástor, Stambaugh, and Taylor (2022) link green assets' realized outperformance to unexpectedly strengthening climate concerns. Ardia et al. (2023) construct a daily newspaper-based climate-concern index and show that green stocks outperform brown stocks on days when climate concerns rise unexpectedly. Their daily stock-return regressions also associate green outperformance with shocks to concerns about specific physical risks, including hurricanes and floods.

I further contribute to research on institutional investors' responses to climate-related shocks. Prior work documents reduced exposure to disaster-zone stocks, greater support for environmental proposals, and portfolio responses to climate-related belief shocks (Alok, Kumar, and Wermers, 2020; Fich and Xu, 2025; Alekseev et al., 2024).[5] I link these findings by examining how the direction of portfolio adjustment varies with political orientation and the extent to which local narratives attribute a disaster to climate change. Holding disaster attention fixed, I show how stronger attribution is associated with active reallocation toward greener assets.

Surveys and portfolio holdings. Finally, the paper relates to research linking investors' beliefs and preferences to observed portfolio holdings (Riedl and Smeets, 2017; Giglio et al., 2025; Anderson and Robinson, 2026). While I don't link individual survey responses directly to their portfolio holdings, I contribute to this evidence on cross-sectional heterogeneity in green ownership by examining portfolio adjustments around particular disaster experiences instead, using pre-event allocations to distinguish investment responses from pre-existing differences in holdings.[6] My results are particularly relevant given standard predictions of portfolio inertia: changes in beliefs explain little about whether investors trade, even though they help explain the direction and magnitude of trading conditional on adjustment (Giglio et al., 2021). The reallocations I observe suggest that disaster experience and its local interpretation can overcome the frictions that ordinarily limit portfolio adjustment. I provide evidence on the experiences and local interpretations associated with active reallocation toward green assets.

2 Data and Measurement

This section describes the data (Section 2.1) and the climate interpretation measure constructed to capture the local information environment (Section 2.2). As outcomes, I use data on household, mutual fund manager, and other institutional investor portfolio holdings, which I combine with records of climate disasters and local newspaper coverage of each disaster to capture how disasters are interpreted locally.

2.1 Data

Households: local investment advisers. To represent household portfolio decisions, I use the portfolios reported by local U.S. registered investment advisers (RIAs) that predominantly serve household accounts. Advisers file Form ADV with the SEC, where they report their client composition, assets under management, and office locations.I combine quarterly Form ADV filings with raw Form 13F holdings to obtain the location and the clientele of each 13F filer. Following Pan et al. [2025], I define an adviser as serving predominantly household accounts when individuals and high-net-worth individuals make up a strict majority of its clients, and as local when all its offices are in one county.[7] The final sample contains 4,040 advisers in 490 counties,

[5] Alok, Kumar, and Wermers (2020) find that funds headquartered within 100 miles of a disaster-zone firm reduce their portfolio weight in that firm more than distant funds, with evidence consistent with salience bias. Alekseev et al. (2024) use local extreme heat and changes in shareholder disclosures to identify climate-belief shocks and connect them to industry reallocations, like buying automobile and semiconductor stocks and selling real estate. Fich and Xu (2025) find increased support for environmental proposals after hurricanes and increased holdings in firms where their votes could help environmental proposals pass, providing evidence consistent with both managers' own preferences and career concerns about investor flows.

[6] Closest to my paper, Anderson and Robinson (2026), study climate-belief revisions following wildfires and global warming attention, and link it to cross-sectional differences in allocations to fossil-fuel-exclusion mutual funds within Sweden's premium pension system, measured about three and a half years after the wildfires. Instead, I compare portfolio active changes from pre-event allocations, tracing the timing and subsequent path of portfolio adjustments over shorter horizons around disasters.

[7] Appendix A.7 reports the construction and selection details.

covering all 50 states, the District of Columbia, and Puerto Rico, from 2009 to 2024 (Table 1). Appendix Table A.6 shows that FactSet classifies most of these advisers as investment advisers (66.4 percent) or as private banking and wealth-management firms (30.6 percent).

Table 1: Registered Investment Adviser Sample Characteristics, 2009–2024

Statistic	All registered advisers	Household-serving	Household-serving, single-county
Registered investment advisers	8,489	4,439	4,040
Adviser-quarters	222,832	111,373	87,853
U.S. counties represented	1,439	850	490
States and U.S. jurisdictions represented	52	52	52
Median RAUM ($ millions)	593.8	468.0	426.1
Median client accounts	439	745	681
Median advisory offices	10.0	8.0	6.0
Advisers ever holding any ETF (%)	87.1	97.3	97.1
Advisers ever holding a green ETF (%)	36.3	45.6	44.0

Notes. This table reports characteristics of the registered investment advisers in the sample over 2009–2024. The universe comprises investment advisers with a direct Form ADV filing or identified as relying advisers on Schedule R; exempt reporting advisers are excluded. Advisers serving predominantly household accounts have a strict majority of individual clients. Single-county advisers report offices in one county. Regulatory assets under management, client accounts, and offices are measured at each adviser's latest observation. ETF indicators require a positive reported position. U.S. jurisdictions include the 50 states, the District of Columbia, and Puerto Rico.

Registered advisers held 41.1 percent of the $10.3 trillion U.S. ETF market at the end of 2024 and 45.2 percent of its green segment, against 20.3 and 17.9 percent for every other Form 13F institution combined.[8] ETFs use is concentrated exactly in the group of advisers serving households. Of the $38.2 billion of green ETFs held by all registered advisers at the end of 2024, advisers serving predominantly household accounts held $18.8 billion, and local household advisers alone held $11.7 billion. The final sample of local household-serving RIAs holds 7.2 percent of the regulatory assets reported by all registered advisers, while holding 20.9 percent of their ETF dollars and 30.7 percent of their green ETF dollars. More than 97 percent of these advisers have ever held an ETF, and more than 44 percent have held a green ETF.

I aggregate 13F filers into stable adviser-level portfolio units,[9] assign each unit to a county, and aggregate positions to the county-quarter level. I restrict the sample to advisers that never change county, so that within-county portfolio changes are not mechanically driven by relocations.

I aggregate ETF holdings across local advisers to measure the allocation of the combined ETF wealth of the households they serve. Under the assumption that these households reside in their advisers' county, pooling holdings gives each household's ETF investment its corresponding weight in the county portfolio. Advisers serving more households or managing larger household investments therefore contribute more to the aggregate. The county green share is

$$g_{ct} = \frac{\sum_{a \in c} \sum_f V_{aft} G_f}{\sum_{a \in c} V_{at}^{ETF}} = \sum_{a \in c} \frac{V_{at}^{ETF}}{\sum_{b \in c} V_{bt}^{ETF}} g_{at}, \tag{1}$$

[8] Appendix Table C.1 shows ETF adoption, portfolio allocation, and market shares for registered advisers and for the household-serving group.

[9] The FactSet parent rollup and the S34 manager identity define the candidate aggregations. I split a parent into separate portfolios only when they belong to distinct advisers and are reported separately throughout. This keeps a portfolio's holdings unchanged when it is reported under a different filer identity. Appendix A.8 reports the criteria.

where G_f identifies green ETFs, V_{aft} denotes adviser a's holdings in ETF f, $V_{at}^{ETF} = \sum_f V_{aft}$, and g_{at} is the adviser's green ETF share.[10]

ESG labels influence investor demand beyond differences in underlying holdings [Baker et al., 2024]. I therefore define an ETF as green if its name in CRSP contains at least one of a list of climate and transition keywords, or if Morningstar assigns it one of a selected set of climate and clean-technology themes. I exclude sectors whose climate mandate is ambiguous[11] and funds with only a social mandate. The rule classifies 227 funds as green over 2009–2024, and Appendix A reports the full classification. Since green ETFs became common in adviser portfolios only around 2015, the main analysis covers 2015–2024 (Section 3.1).[12] At the end of 2024, 179 green ETFs were active, with a market value of $94.45 billion, or 0.92 percent of the U.S. ETF market.

To ensure that an adviser's county is also where its clients live, I require the principal office and every branch office to be in the same county, and I manually review the names of all retained advisers. I exclude internet-based advisers and advisers whose offices are all in one county but whose clientele is not local. To validate the local definition, I verify that the share of an adviser's stock portfolio invested in firms headquartered in its own state exceeds the share obtained by randomly reassigning adviser locations. I compute the green share within an adviser's ETF holdings, so that changes in mutual fund holdings reported on the same Form 13F do not enter the household measure.[13]

Finally, I investigate who the households behind these portfolios are. ETFs are held largely by households, directly or through their advisers. Advisers whose clients are mostly high-net-worth individuals held 46.9 percent of their reported 13F assets in ETFs by 2023, against 6.6 percent for pensions and endowments [Moussawi et al., 2025]. In 2024, 16.9 million U.S. households, 13 percent of the total, owned ETFs. These households are relatively wealthy, with a median income of $140,000 and median financial assets of $500,000 [Investment Company Institute, 2025], against a national median household income of $87,460 in 2025 [U.S. Census Bureau, 2026]. Since 58 percent of U.S. families hold equity directly or indirectly [Board of Governors of the Federal Reserve System, 2023], more than one in five investing households owns ETFs. These facts suggest that the local and household-account restrictions keep the part of an adviser's 13F portfolio that most likely reflects its clients' choices, especially for green ETFs.

Mutual fund managers. I combine fund holdings from the Thomson Reuters S12 database, fund characteristics and manager names from Morningstar Direct, manager profiles from Revelio Lab., and the filings of the managing firms with the SEC to build a panel of fund portfolios located where their managers work. Since mutual fund investors cannot be located, I study the portfolio managers who actively rebalance the funds, whose location I observe. I locate each fund at the offices of its sub-adviser, or of its adviser when it

[10] Adviser-level regressions weighted by pre-disaster shares of county ETF holdings approximate this aggregate response and yield very close estimates, as reported in Appendix Table E.11. They capture changes in green shares within advisers but omit the contribution of changes in advisers' relative ETF holdings; Online Appendix OA.A derives the conditions for equivalence. Equal-adviser-weighted regressions instead measure the average adviser's response, giving advisers equal importance regardless of the number of households they serve or the ETF wealth they manage. Although this differs from my target, I report these estimates in the Appendix: the results remain statistically significant, while their magnitudes change with the different estimand.

[11] Metals and mining, resource extraction, upstream and midstream energy, commodities, agriculture and food, nuclear, water, and batteries.

[12] Green ETF market value grew from $2.0 billion in 2009 to a peak of $105.6 billion in 2021.

[13] Some advisers also manage mutual funds, and their 13F filings then include the funds' holdings. This overlap matters little for ETF positions. In the same quarter, mutual funds in S12 hold only 1.23 percent of ETF market value and 2.11 percent of green ETF market value, and only 0.59 percent of S12 funds hold any green ETF. The Appendix shows that the results are similar for flows normalized by lagged assets and for dollar flows.

has none, from Form N-SAR until 2017 and Form N-CEN thereafter, following Alekseev et al. [forthcoming], and from Morningstar portfolio-manager records. This office often differs from the fund's headquarters (Section 3.2.2). I also match 1,161 funds to 2,348 named managers by combining Morningstar Direct and Revelio, which allows me to measure the managers' political views from their political donations. I construct fund-quarter holdings from S12 and compute changes only between consecutive quarterly reports. I also restrict the sample to actively managed U.S. equity funds.[14] Following Sammon and Shim [2023], I decompose these changes into a component implied by fund flows and an active rebalancing component. To measure the environmental profile of the stocks the funds trade, I use the Environmental Pillar score from LSEG [LSEG, 2024], the part of the ESG rating related to climate change. Appendix Figure A.4 maps the county-weighted geography of fund management.

I classify a mutual fund as having an ESG mandate if its name includes a sustainability term or if Morningstar classifies it as having an intentional ESG mandate. I exclude funds with an ESG mandate from the analysis of managers (Section 3.2), because their portfolios follow climate considerations and incentives unrelated to local disasters [Alekseev et al., forthcoming]. I retain Low Carbon-only and unlabelled funds in the manager sample. I also compare flows into ESG and confirmed non-ESG funds at the end of Section 3.1, using a classification that additionally includes Morningstar climate-theme funds and excludes Low Carbon-only and unlabelled funds.

Other institutional investors. The remaining 13F filers are mostly investment advisers and investment companies serving firms, municipalities, pension funds, and other institutions, together with hedge funds, banks, and insurers. I locate hedge fund managers, and other advisers whose clients or decisions appear local (e.g., those serving firms or municipalities), with the same rules as for the household advisers, applied to their Form ADV filings under full registration or as exempt reporting advisers. I map 1,561 hedge fund managers to 117 counties and 6,757 other registered advisers to 1,438 counties, counting every office. Following Baik et al. [2010], I place pension funds, banks, and insurers at the county of their headquarters, where their portfolios are managed centrally.

Climate disaster episodes. I build disaster data from the NOAA Storm Events Database, which records storms and other weather events at the county level. I group related county records into disaster episodes, so that a disaster that hits several counties counts as a single event while each county keeps its own exposure.[15] I assign each episode a dominant hazard type and measure its severity from inflation-adjusted property and crop losses, fatalities, and injuries. I combine these measures into an index of the episode's aggregate impact.[16] For each sample, I retain the episodes occurring in the period of analysis that are in the top 15 percent of this index, within or across hazard types, and exclude episodes that had another local episode in their pre-period. Appendix Table A.1 reports the characteristics of the episodes by hazard type, and Figures C.1 to C.2 map them by county.

[14] I exclude funds whose Thomson Reuters objective is international, municipal bonds, bond and preferred, or balanced [Alok et al., 2020, Kacperczyk et al., 2008], passive funds and exchange-traded funds, identified by CRSP flags and fund names [Appel et al., 2016, Sammon and Shim, 2023, Alok et al., 2020], and funds with less than $5 million in assets or fewer than ten stocks [Alok et al., 2020].

[15] I audit every multi-event episode for coherence and split those that fail into sub-episodes, so that each episode approximates the footprint of one physical shock (Appendix A).

[16] The construction of this index builds on a Pareto approximation of the tail of disaster losses **[MODEL OF RARE DISASTER]**. Appendix A provides additional details on the construction.

Local news. To measure how disasters are discussed locally, I collect newspaper articles from ProQuest TDM Studio. I query articles whose text includes climate and global-warming vocabulary, as well as extreme-weather and disaster vocabulary, yielding 1,485,896 disaster articles from 527 outlets. Restricting to articles that cover a disaster over 2015–2024, the corpus holds 782,714 articles, of which 724,969 come from local outlets and 57,745 from national ones. I combine the articles with local newspaper circulation data from the Alliance for Audited Media and Editor & Publisher, complemented by device-measured readership from Comscore. The circulation data give ZIP-level newspaper market shares, which I use to aggregate outlet-level measures to the ZIP and county level. Using these weights, I construct a sample of 433 outlets reaching 2,134 counties. Appendix A provides additional details on the construction of the corpus and of the circulation weights.

Additional data. To construct the instrument for local interpretation (Section 5), I combine newspaper ownership data from [Ewing, 2026] with the ownership panels of the Northwestern Medill Local News Initiative. I date ownership changes to the quarter using SEC Form 8-K filings and reports of newspaper and company acquisitions. The remaining sources enter as controls, such as characteristics of ZIP codes, counties, advisers, mutual funds, institutional investors, and disasters, or serve to construct placebo events and other newsworthy events that coincide with disasters. Appendix A describes the source and construction of each of these datasets.

2.2 Local Information and Interpretation Environment

Climate beliefs are shaped by local discourse, through family and friends, news from different sources, and social media. Thirty-nine percent of Americans hear about global warming in the media at least once a month, and 17 percent hear people they know talk about it as often [Leiserowitz et al., 2026]. Local news reaches people through the same channels. Seventy-three percent of Americans get local news from friends, family, and neighbors, 64 percent from local television, half from local social media groups or forums, and a third directly from local daily newspapers, down from 43 percent in 2018 [Pew Research Center, 2024b,a]. I therefore measure how much local newspapers cover each disaster and how they interpret it. These two measures aim to capture the narratives shared locally and the local discussions they trigger.

Although only a third of residents read a local daily directly, newspapers shape much of what circulates locally. First, they originate most of the stories that set the agenda of other media,[17] and even small outlets can spark wider discussion [King et al., 2017]. Second, their articles reach readers beyond print, through websites, news aggregators, and social media.[18] Finally, local newspapers also reach investors. Retail and institutional investors rely on geographically proximate information [Engelberg and Parsons, 2011, Baik et al., 2010, Bernile et al., 2015], and local newspapers are "a key channel through which institutional investors acquire geographically scattered information" [Kang and Nam, 2025].

To construct a measure of local climate interpretation, I classify every paragraph of the disaster articles with a hierarchical teacher–student pipeline of large language models. I first define the classes and hand-label about 300 paragraphs, which I use to tune the prompts of a teacher model, GPT-5.5. The teacher

[17] Pew Research Center, *How News Happens*, 2010, https://www.pewresearch.org/journalism/2010/01/11/how-news-happens/.

[18] Of the 546 local newspapers in the sample, at least 468 have an active website and 293 a Twitter account. Within a county, their print market shares track their web market shares (correlation 0.78; Appendix Figure D.5), and their Twitter audiences track their print circulation (rank correlation 0.84; Appendix Figure D.4). News aggregators matter most for the audiences of small and local outlets [Calzada and Gil, 2020, Athey et al., 2021].

then labels a training set, on which I train two local student models, DeBERTa and Qwen. DeBERTa classifies the full corpus and passes uncertain cases to Qwen. The classifier first detects whether a paragraph discusses a specific disaster event, identified in time and place. If it does, the classifier records whether the paragraph links the event to climate change and how strongly, and its stance on climate change. The stance can be affirming, denying, neutral, lukewarm (accepting global warming but minimizing its importance), or reporting contested opinions, and the classifier also records whether the stance is the reporter's own. In addition, I identify other common topics in each paragraph, the most frequent being damages and the human toll (damage to homes and victims). Against my own hand labels on held-out paragraphs, the two student models reach an F_1 score of 0.87 and 0.88 on the climate link, above general-purpose models such as gpt-4.1-mini (0.85) and gpt-5.4-mini (0.77).[19] Appendix Table A.4 shows two examples. The first, from the coverage of Hurricane Helene, links the storm to the burning of fossil fuels without ever mentioning "climate change," so a keyword search would miss it, while the classifier labels it as an affirming climate link. The second, from the coverage of Hurricane Katrina, tells the human toll through one person's story and makes no link to climate change.

The classifier yields two measures, the first one being local disaster coverage. Let o index outlets, c counties, and e disaster episodes, and let w_{co} be outlet o's circulation share in county c, normalized to sum to one within the county, and A_{oe} the number of articles on episode e that outlet o publishes in the four weeks from the episode's start. Local disaster coverage of episode e in county c is the circulation-weighted number of articles,

$$A_{ce} = \sum_{o} w_{co} A_{oe}. \tag{2}$$

When coverage enters the regressions, I use $\log(1 + A_{ce})$ to allow for diminishing returns to additional coverage.

The second measure is local climate interpretation, the extent to which local coverage of a disaster interprets the event as a result of climate change. Let C_{oe} be the number of outlet o's articles on episode e that link the event to climate change, and $m_{oe} = C_{oe}/A_{oe}$ the outlet's interpretation ratio, defined for outlets with $A_{oe} > 0$. I aggregate these ratios to the county with the same circulation weights, renormalized over the outlets that cover the episode

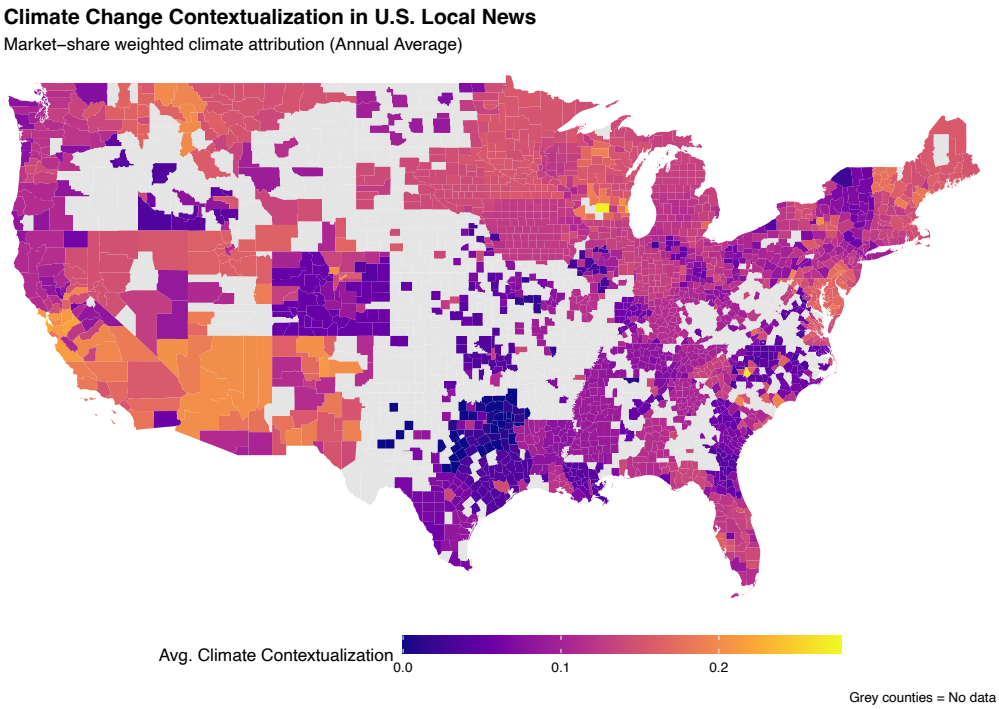
$$M_{ce} = \frac{\sum_o w_{co} \mathbf{1}\{A_{oe} > 0\} m_{oe}}{\sum_o w_{co} \mathbf{1}\{A_{oe} > 0\}}. \tag{3}$$

M_{ce} is the share of climate-linked articles in the coverage of the average local outlet that covers the episode, with outlets weighted by their readership in the county. Each outlet counts in proportion to its readership, whatever the number of articles it publishes on the episode.[20] The measure averages 0.172 across county–episode observations, with a standard deviation of 0.157, so an increase of 10 percentage points corresponds to 0.64 standard deviations. In a county served by a single outlet, such an increase means one more climate-linked article in every ten articles on the episode. In a county served by two outlets with equal circulation, it means either one more climate-linked article in ten in both outlets, or two more in ten in one

[19] Appendix Figure A.1 shows the pipeline. Appendix Tables A.2 and A.3 report the performance of each model on every classification task, and Appendix A details the training and validation procedures. Appendix A also reports the full set of classes, including paragraphs on disasters in general, climate change, climate policy, and peripheral climate topics.

[20] As a robustness check, I also use the pooled share $\sum_o w_{co} C_{oe} / \sum_o w_{co} A_{oe}$, which weights outlets by both their readership and the number of articles they publish on the episode.

FIGURE 1: LOCAL CLIMATE INTERPRETATION OF DISASTER COVERAGE



Notes. Mean local climate interpretation of disaster coverage in each county, M_{ce} in Equation (3), over the disasters covered by its local outlets. Grey counties have no local coverage of disasters in the sample.

outlet and no change in the other.

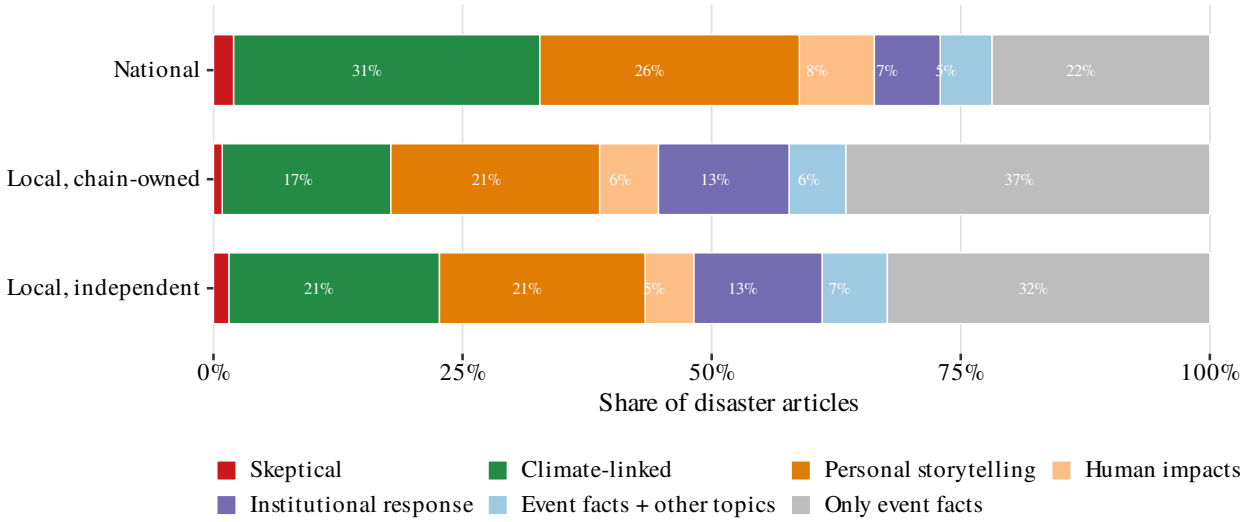
Figure 1 shows the average local climate interpretation of disaster coverage in each county. Interpretation varies widely across the country: it is highest along the West Coast and in the Southwest, and lowest in Texas and much of the South. It also rises over time, from 6.5 percent of a county's local disaster coverage before 2015 to 17.9 percent from 2015 on.

To separate the role of the outlet from that of the disaster, I decompose the variance of coverage and of climate interpretation across outlet–episode pairs into an outlet component, an episode component, and their match, as in the literature on matched employer–employee data [Abowd et al., 1999, Kline et al., 2020]. The outlet component captures stable differences across outlets, such as the political views and green attitudes of their audience and owner. The episode component captures the characteristics of the disaster, such as its year, hazard type, and damages. The match captures how a given outlet covers a given disaster beyond what the outlet and the disaster predict on their own. Appendix Figure D.3 shows that coverage depends mostly on the episode, which accounts for 55.4 percent of its variance against 19.5 percent for the outlet, mainly through the hazard type.

Climate interpretation instead lies mostly in the match, at 54.1 percent of its variance. The event component accounts for 40.7 percent of the variance in climate interpretation. Over 90 percent of its explained part comes from the year of the event, because the interpretation of each hazard type changes over time. Stable outlet and audience differences, together with common event characteristics, including time, still leave more than half of the variation unexplained. A substantial part of the variation therefore concerns how a specific outlet interprets a specific disaster, while serving a given audience at that time. Part of the match reflects how articles reach an outlet. Many local newspapers belong to chains, groups of newspapers

under the same owner, and a chain often writes an article once and prints it in several of its newspapers. Among the outlet-event pair characteristics I observe, chain copies account for 92 percent of the explained part of the match. I leverage this variation to provide causal evidence on the effect of local interpretation in Section 5.

FIGURE 2: COMPOSITION OF DISASTER COVERAGE BY OUTLET OWNERSHIP



Notes. This figure plots the composition of disaster coverage for national outlets, chain-owned local outlets and independent local outlets. Shares are calculated among 424,888 disaster articles published from 2015 through 2024: 41,016 national, 353,748 chain-owned local and 30,124 independent local. A local outlet is chain-owned when its owner holds two or more newspaper titles in the quarter of publication, and independent when its owner holds one. Categories are mutually exclusive. Skeptical treatment takes precedence over climate linkage, and climate linkage over the narrative and contextual categories. An article is assigned to Only event facts when every relevant block reports the specific event. It is assigned to Event facts + other topics when at least one relevant block is not about that event. This block either discusses hazards as a class, a trend or a projection, or covers green products, sustainability or climate regulation.

The way a disaster is represented in the news therefore depends on the type of outlet that covers it. Figure 2 shows the composition of disaster coverage for national outlets, chain-owned local outlets, and independent local outlets. The key difference is in the share of disaster articles linked to climate change, which is 30.8 percent for national outlets, 16.9 percent for chain-owned local outlets, and 21.1 percent for independent local outlets. Across all disaster articles, chain-owned local outlets link disasters to climate change 4.2 percentage points less often than independent ones, and the gap remains statistically significant within the same quarter (Appendix Table D.1).

I interpret this measure as an equilibrium local interpretive environment: it captures whether local discussions and narratives, proxied by newspaper coverage and framing, map a realized disaster into climate meaning. Section 5 will isolate the effect of one source of exogenous supply-driven variation in this measure coming from shocks in newspaper ownership structures to leverage the variation in local interpretation described above.

3 Local Responses to Interpretation

Households respond systematically to the climate interpretation of disaster coverage by raising the green share of their portfolios, while coverage with lower or absent climate contextualization causes turnover without net buying into sustainability. The response decays steeply with distance and concentrates among the directly affected, consistent with experiential updating, and parallel analyses of household climate beliefs and climate-related behavior show the same post-disaster and interpretation-dependent pattern. Non-green mutual-fund managers show the same separation, with salience and interpretation moving on separate lanes. Managers in affected counties have a more negative flow-adjusted response toward disaster-zone firms than far managers do, regardless of interpretation, consistent with the divestment documented by Alok et al. [2020]. Nearby managers instead respond to the local interpretation, tilting their active adjustment toward greener firms when local coverage contextualizes the disaster as climate change.

3.1 Households

More climate-interpreted coverage moves the local portfolio toward green funds over the two quarters after the disaster, on both the extensive and the intensive margin; in counties without significant climate-linked coverage, disaster coverage moves portfolio turnover without a corresponding net tilt toward sustainable funds; absent disaster experience, the interpretation effect strongly decays.

The empirical analysis is built around repeated disasters in a county-quarter panel. I rely on the local-projection difference-in-differences of Dube et al. [2025], on its horizon-specific clean-control logic and on its extension to non-absorbing treatment. Appendix B reports the horizon-by-horizon clean-window rules, and related robustness checks. Identification requires that, absent the episode, the green share of affected and unaffected counties linked to it would have moved in parallel; conditioning on the adjacent prior change in the outcome relaxes that to conditional parallel trends.

In the main analysis, I restrict to flood episodes, forcing the comparison to always be between events of the same hazard type.[21] Floods are the events with more geographical span, especially covering counties where RIAs are located. (See Figure C.2.) Green ETFs started being more diffused as a means of sustainability only since the 2010s, and became relatively more present in investment adviser portfolios around 2018. Therefore, in my main analysis I focus on 2018, to reduce concerns for the estimate capturing trends in availability of these instruments and the trend in national attention to climate change.[22]

The outcome is the share of a county's exchange-traded fund holdings in environmentally focused funds, measured through the household-serving local advisers of Section 2 and aggregated to the county of the adviser's office. The main text reports the two-quarter cumulative fixed-price change, alongside the current-price change,[23] rescaled to basis points.[24]

To isolate active rebalancing from valuation drift, I construct the fixed-price change: for each episode starting in quarter t and each horizon h, I rebuild the county portfolio using quantities observed at $t + h$,

[21] Results are robust to extensions considering different sets of hazard types, once exposure and the interpretation response are properly allowed to vary by event type (disaster-zone × hazard fixed effects). In particular: (i) a group of hazards with a comparable level of scientific attribution certainty (floods, hurricanes and wildfires); (ii) all attributable climate hazards (FHW as well as slow-onset but strongly attributable events such as heatwaves and droughts); (iii) all climate hazards with different levels of attributability and certainty (e.g., including tornadoes, hail, winter storms).

[22] The appendix presents analogous and comparable results for 2010 onwards as well.

[23] The fixed-price measure isolates active reallocation, while the total measure captures the net change in observed green exposure.

[24] Results are robust to non-cumulative outcomes, alternative post-event windows, and baseline definitions that average across pre-event quarters. I use the single-quarter baseline to avoid mixing the reference point with earlier pre-trend movement.

valuing every ETF at the pre-event price vector $P_{j,t-1}$,

$$s^{FP}_{c,t+h|t-1} = \frac{\sum_j g_j P_{j,t-1} Q_{cj,t+h}}{\sum_j P_{j,t-1} Q_{cj,t+h}}. \tag{4}$$

The corresponding LP dependent variable is the cumulative level change $\Delta_h s^{FP}_{ct} \equiv s^{FP}_{c,t+h|t-1} - s^{FP}_{c,t-1|t-1}$; for the observed-price share, $\Delta_h s_{ct} \equiv s_{c,t+h} - s_{c,t-1}$. For unit i, a county or an adviser, disaster episode e starting in quarter t, and horizon h, I estimate

$$\Delta_h Y_{ie} = \alpha^h_i + \lambda^h_t + \eta^h_e + \beta_h D_{ie} + \delta^h_{y(e)} D_{ie} + \zeta^h_{k(e)} D_{ie} + \theta_h D_{ie} \times M_{ie} + \Gamma'_h X_{i,t-1} + \varepsilon_{ieh}, \tag{5}$$

on a sample of units in the disaster zone of the episode at t and of clean controls, units linked to the episode, outside the disaster zone, and with no other disaster over the horizon. I restrict affected units to those whose two quarters before the start have no earlier local disaster. D_{ie} is an indicator function and takes value 1 when i is in the disaster zone of episode e, while M_{ie} is the climate-interpreted share of local coverage of the episode;[25] α^h_i are unit fixed effects; λ^h_t calendar-quarter fixed effects; η^h_e episode fixed effects, absorbing the level of each disaster; β_h the affected-unit effect; $\delta^h_{y(e)}$ year-specific affected-unit effects, capturing the disaster-zone response common to a year; and $\zeta^h_{k(e)}$ event-type affected-unit effects, capturing the response common to a hazard type, present when the sample spans more than one hazard.

$X_{i,t-1}$ is a vector of county controls that may be correlated with both local interpretation and the portfolio response. It always includes the coverage and interpretation main effects, with additional controls and fixed effects depending on the analysis. The episode fixed effects compare units in the disaster zone with units linked to the same disaster. The year-specific and event-type-specific affected-unit effects compare the interpretation effect among affected units (the slope) within the same year and the same hazard type. I cluster standard errors at the county and episode level. The main estimates are reported at $h = +1$.

Interpretation Effect on Green Household Demand. The full results are presented in Table 2. I find that a one-standard-deviation increase in local climate interpretation of the disaster raises the pre-event-price green share of affected counties by 30.31 basis points relative to the unaffected counties linked to the episode.[26] At observed prices the estimate is 29.50 basis points, within one basis point of the fixed-price estimate. This proves that the increase is not confounded by price drift in either direction. I can exclude both that advisers' increase in relative exposure is the artifact of simple revaluation (since active rebalancing is equally high), and that active re-allocation has been a counterbalancing response to negative drift to maintain desired exposure. Therefore, this change reflects an increase in *desired* allocation.

Of two counties linked to the same disaster, the one whose local coverage connects the event to climate change ten points more often, thirty percent of articles in place of twenty, holds 19.30 more basis points of green share at fixed prices two quarters later. That corresponds to $0.66 interpretation-induced extra millions reallocated to green ETFs by the median county-quarter ETF holdings of these advisers ($340.3 million), and $5.04 millions at the mean holdings ($2.61 billion). The two-quarters cumulative change induced by 10 percentage point share belongs to the 82nd percentile of all absolute two-quarter changes on this support. Results are robust to different specifications of this design.[27]

[25] M_{ie} is missing for county-episodes with no local coverage, so the interaction is estimated on covered county-episodes.

[26] The estimate is column 4 of Table 2, with all controls and their interactions with D_{ie}. Standard errors are clustered by county, the more conservative choice: $p = .017$, against $p = .009$ when clustering by episode. With county fixed effects the estimate is 22.76 ($p = .068$); with state-by-year fixed effects, 20.40 ($p = .213$).

[27] For example, robust to extensions of the non-overlapping window to concurrent-episode weighting functions (see Appendix B

for their constructions). Moreover, Appendix Table E.1 reports the same control columns with interpretation measured over the calendar quarter, and Table E.2 reports the same estimation at the adviser level. The comparison does not rest on the control counties: comparable differential when estimating the slope among the affected counties alone, while the same contrast among unaffected counties is indistinguishable from zero (Appendix E, Figure E.5).

Table 2: County green-ETF portfolio response

	Dependent variable:									
	Green-ETF share response (basis points)									
	Pre-event prices					Current prices				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Affected × Interpretation	25.48**	25.86**	22.46*	30.31**	22.76*	25.87**	25.58**	22.53*	29.50**	22.55*
	(10.13)	(12.33)	(12.18)	(12.58)	(12.44)	(10.46)	(12.50)	(12.25)	(12.62)	(12.28)
Affected × Attention	.86	4.91	3.64	1.69	−9.36	.46	4.55	3.40	1.79	−9.16
	(10.93)	(10.92)	(10.30)	(10.06)	(10.01)	(11.17)	(10.84)	(10.24)	(10.08)	(10.11)
Total Interpretation effect among affected counties	26.70**	27.08**	23.47*	31.32**	22.73*	27.02**	26.73**	23.43*	30.43**	22.54*
	(10.45)	(12.44)	(12.26)	(12.53)	(12.31)	(10.75)	(12.61)	(12.32)	(12.58)	(12.16)
Observations	105,662	105,662	105,662	105,662	105,662	105,662	105,662	105,662	105,662	105,662
R^2	.028	.029	.031	.032	.102	.030	.030	.033	.033	.106
Affected county-episodes	244	244	244	244	244	244	244	244	244	244
Episode and quarter FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
State FE	✓	✓	✓	✓		✓	✓	✓	✓	
County FE					✓					✓
Affected × hazard × year FE		✓	✓	✓	✓		✓	✓	✓	✓
Financial and demographic controls			✓	✓	✓			✓	✓	✓
Own-county damage, deaths and injuries			✓	✓	✓			✓	✓	✓
Political and belief controls				✓	✓				✓	✓
Lagged climate attention				✓	✓				✓	✓
Affected × controls				✓	✓				✓	✓

Notes. This table reports estimates of the county green-ETF share response to disasters. The outcome is the average county green-ETF share over the quarter of the episode and the two following quarters minus its average over the two quarters before, in basis points, at pre-event prices, held at their level in the quarter before the episode (columns 1–5), and at current prices (columns 6–10). The sample covers high-salience flood and coastal episodes, 2015–2024. Affected is one for counties in the episode footprint. Interpretation is the climate-interpreted share of local coverage of the episode and Attention the log volume of that coverage, both over the first four weeks of the episode and standardized; one standard deviation of Interpretation is 15.7 percentage points. Each regression also includes Interpretation and Attention and, in columns 1 and 6, Affected; their coefficients are not reported. The total Interpretation effect among affected counties is the sum of the Interpretation coefficient and its Affected interaction, with its standard error from their joint covariance. Financial controls are lagged log portfolio assets; demographic controls are income, college share, white share, population and population density; political and belief controls are the prior Democratic vote share and the county share worried about global warming; lagged climate attention is the share of local coverage on climate policy and green topics in the quarter before the episode. Observations are weighted by the inverse number of episodes starting in the quarter. Standard errors, in parentheses, are clustered by county. A nonparametric bootstrap that resamples counties with replacement (999 draws) gives $p = .036$ for Affected × Interpretation in column 4 and $p = .032$ in column 9. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

Entry and Incumbent Weight Both Move. Per standard deviation of interpretation in affected counties, the share of advisers in the county holding a green ETF rises 5.76 percentage points, and 5.94 among advisers reporting from the same county across the two quarters, so the increase comes from advisers already there. Over the same two quarters that share rises 1.35 percentage points on average across the county-episodes of this sample, so the interpretation response is more than four times the ordinary post-disaster movement. Among advisers in the disaster zone, the response is 18.6 basis points per ten percentage points of interpretation, which is $223,998 at the median adviser ETF book and $446,069 at the mean.[28]

Coverage Moves Size, Interpretation Moves Direction. Gross turnover of the whole reported book rises with disaster coverage in affected counties, by 0.201 log points in the quarter after the event, while climate interpretation itself adds nothing (Table E.12). The differential portfolio response to interpretation therefore validates that attention and conditional interpretation (here captured by disaster experience, coverage and its newspaper interpretation) are separate independent constructs by moving different margins of the same portfolio.

Coverage and interpretation could simply move together across counties, with more heavily affected places both covered more and interpreted more, so a reader could take the interpretation effect for the disaster's own attention effect showing through a correlated measure. The turnover result does not fit that reading, its coverage response holding whether or not green securities are counted while its interpretation coefficient stays null. Attention, measured by how much a county trades, moves the portfolio's size; only interpretation, holding coverage fixed, moves its direction toward sustainability.

The Response Strongly Decays With Distance. Local climate interpretation induces portfolio changes toward sustainability where the disaster is lived; as I will discuss in Section 4, local coverage of non-experienced disasters does not make investment-advised households' portfolios greener, regardless of local narratives or interpretations.

[28] See Table E.4.

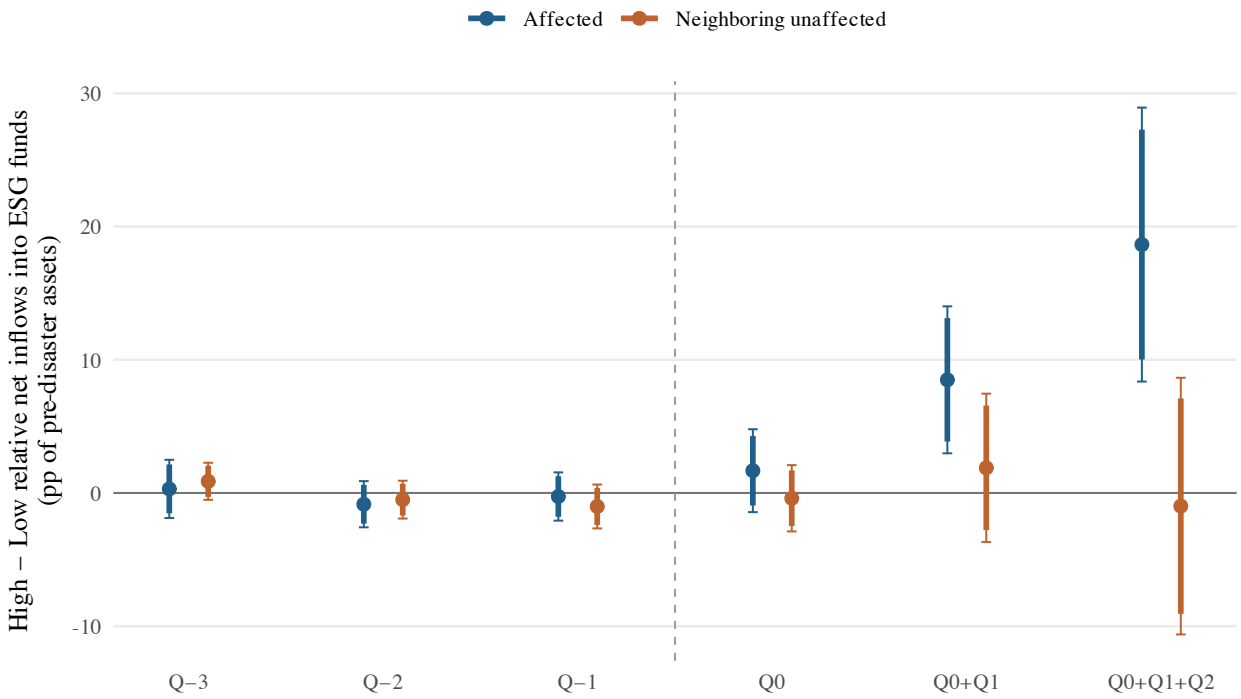
Extension to ESG Mutual-Fund Flows. The main analysis uses geographically detailed ETF holdings of household-serving advisers to relate local climate interpretation to portfolio allocation. To establish external validity beyond green ETFs, I extend the analysis to ESG mutual-fund flows.[29] Disaster exposure is assigned at the fund's headquarters state, assuming that headquarters location proxies for clientele exposure, motivated by evidence of geographic familiarity in mutual-fund selection [Bailey et al., 2011]. Using this approximation, Marshall et al. [2021] find higher net inflows into green mutual funds relative to matched nongreen funds when disasters occur in the funds' headquarters states. In Figure 3 I compare relative net ESG inflows, measured against net inflows into non-ESG funds, across disasters with high and low climate interpretation, holding disaster attention and other covariates constant.

Fund fixed effects absorb persistent differences in fund flows. ESG-status-by-quarter fixed effects absorb broad national trends in ESG investment by allowing ESG and non-ESG fund flows to vary separately each quarter. Hazard-by-exposure-group fixed effects account for average differences across disaster types and geographic exposure.

Among funds headquartered in disaster-affected states, relative net ESG inflows are larger following high-interpretation disasters, by 8.49 percentage points of pre-disaster assets over two quarters and 18.64 points over three quarters. The corresponding differences in neighboring unaffected states are statistically insignificant. These findings provide suggestive evidence that the interpretation–investment relationship extends from advisers' green ETF allocations to ESG mutual-fund flows.

[29]ESG funds are identified by a dated green or sustainable fund-name classification, a Morningstar intentional-ESG designation effective before the disaster, or a Morningstar climate-theme flag. The comparison group comprises confirmed non-ESG mutual funds; ETFs, low-carbon-only funds, and funds with ambiguous classifications are excluded.

FIGURE 3: CLIMATE INTERPRETATION AND FLOWS INTO ESG MUTUAL FUNDS



Notes. This figure plots the High-minus-Low difference in relative net inflows into ESG mutual funds, measured against non-ESG funds, separately for funds headquartered in disaster-affected and neighboring unaffected states. High and Low interpretation are defined using the median interpretation among disaster-affected state observations, with the same cutoff applied to both groups. The outcome is net dollar flow divided by pre-disaster assets, in percentage points. Pre-event points use individual quarterly flows; post-event points accumulate flows from Q0, with the same asset denominator. Each horizon is estimated separately on fund–disaster observations, 2015–2024. High interpretation and its interaction with ESG status enter separately by band, together with ESG status × Affected. Attention and its ESG interaction also enter separately by band. Fixed effects are fund, ESG status × event-calendar quarter, and hazard × band. Controls are log disaster damage and its ESG-status/band interactions, and log pre-disaster assets. Each supported fund-quarter receives total weight one across episode links. Thin and thick bars are 95% and 90% confidence intervals, clustered by disaster episode. Appendix H states the regression and identifying comparison; Figure H.1 uses continuous interpretation.

3.2 Mutual-Fund Managers and Local Climate Interpretation

The adviser setting of Section 3.1 leaves two gaps. First, households delegate portfolio management to advisers, so changes in green allocations may reflect the advisers' decisions as well as their clients' demand. Second, local interpretation and political views are measured for the county, so they characterize the local information environment and the surrounding population, and not the beliefs of the particular households making the investment decisions. Looking at mutual funds allows me to address both gaps. In this context, I can observe the investment teams that make the decisions and measure the political views of the managers themselves. The hypothesis is that managers learn about climate change from how a nearby disaster is covered where they work. When local coverage interprets the disaster as climate change, nearby managers tilt their portfolios toward greener firms.

3.2.1 Empirical Strategy

I study how the portfolios of these funds change around climate-related disasters from 2015 to 2024. In each quarter, I link a fund to the nearest disaster and to every disaster that hits one of its offices.[30] I then place each fund's offices in distance bands around the disaster, with equal weight on each office. Managers are affected when an office lies in a county hit by the disaster and nearby when it lies within 50 miles of one. Managers more than 100 miles away are the comparison group. A fund with offices in several bands belongs to each in proportion to its offices there. I exclude a fund from the analysis of a disaster when it has an office within 100 miles whose county had no local coverage of the disaster, since local interpretation is not defined there. I also exclude it when stocks with an Environmental Pillar score make up less than half of its stock portfolio before the disaster.[31]

The stocks that mutual funds trade carry no green label like the green ETFs of Section 3.1, so I measure each stock's environmental profile with its Environmental Pillar score (Section 2.1). The outcome is the green tilt of the fund's trading, the change in its weight in stocks in the top tercile of the score minus the change in its weight in stocks in the bottom tercile, over the quarter of the disaster, in basis points of the fund's scored stock portfolio. As for households (Equation 4), I compute the weights at the beginning and at the end of the quarter using prices at the beginning of the quarter. The outcome therefore captures the managers' own trades and is not driven mechanically by changes in prices. I also report the change at current prices. To validate the measure, I use the approach and data of Alekseev et al. [forthcoming], extended to 2023, which treat local heat waves as shocks to managers' climate concerns. A fund with all its offices exposed to such a heat wave raises its green tilt by 65.39 basis points at beginning-of-quarter prices and 71.40 at current prices (Appendix Table E.19).

Isolating the effect of local interpretation on managers has two main challenges. First, how local outlets interpret a disaster may be endogenous and reflect stable characteristics of the place and of the managers who live there, and not the event itself. Namely, counties whose outlets often write about climate change, or with a given political leaning, may also host managers who respond differently to any disaster. Second, the disaster itself can change the stocks the managers hold and the flows into their funds, whatever its coverage. To address the first challenge, I include fixed effects for the county from which the fund is managed, which absorb how funds managed from that county respond to disasters in general. I also control for how the county's outlets interpret the other large disasters they cover, so that the comparison holds fixed the county's usual reading of climate events. To address the second challenge, I include disaster fixed effects to absorb the movements of greener and browner stocks in the quarter of the disaster, which are common to all managers.

Finally, instead of focusing on directly hit counties, where the disaster caused the most damage,[32] I estimate the effect on managers located in counties near those directly affected.

For fund f and episode e, I estimate the following regression

$$\Delta Y_{fe} = \eta_e + \mu_{c(f,e)} + \kappa_{s(f)} + \sum_{b \in \{a,w\}} \beta^b_e D^b_{fe} + \beta^n D^n_{fe} + \sum_b \theta_b D^b_{fe} \times M^b_{fe} + \Gamma' X_{fe} + \varepsilon_{fe}, \quad (6)$$

where M is the share of local coverage that interprets the disaster as climate change (Equation 3). The fund's

[30] As disasters I consider floods, hurricanes, wildfires, severe storms, heat waves, and droughts. I exclude a fund from the analysis of a disaster when one of its offices was hit by another disaster in the same or the previous quarter.

[31] This leads to exclude around 2.7 percent of observations. For the median fund in the final sample, 94 percent of the stock portfolio before the disaster is scored. The final sample comprises 37,047 observations of 2,355 funds around 318 disasters.

[32] Managers in counties directly hit by the disaster also face its direct consequences, such as damage and outflows from local investors.

offices fall into four distance bands. D^a_{fe} is the share of the fund's offices in the counties hit by the disaster, D^w_{fe} the share within 100 miles of them, D^n_{fe} the share within 50 miles, a subset of the previous band, and D^d_{fe} the share more than 100 miles away. M^b_{fe} averages M over the fund's offices in band b. The coefficient of interest is the total effect among nearby managers, $\theta_w + \theta_n$.

η_e are disaster fixed effects that absorb market-wide movements of greener and browner stocks in the quarter of the disaster. β^b_e are band-by-disaster fixed effects for affected managers and those within 100 miles, capturing each group's average response to each disaster, including its damage. β^n is a level common to all disasters for managers within 50 miles. $\mu_{c(f,e)}$ are fixed effects for the county from which the fund is managed, which absorb how funds managed from that county respond to the large disasters in the sample. For a fund with offices in several counties, this is the county with most of its offices, and the one closest to the disaster in case of a tie. $\kappa_{s(f)}$ are fund-style fixed effects, which remove average differences across investment styles. X_{fe} contains the volume of local coverage in each band, the lagged size, age, and turnover of the fund, damage in the counties hit and, depending on the column, county and political controls. Each control enters separately for the fund's offices in every distance band.[33] I weight each observation by one over the number of disasters linked to the fund in the quarter, and I cluster standard errors by disaster.

The effect of interpretation is therefore identified by comparing managers at the same distance from the same disaster whose local coverage interprets it more or less as climate change.

3.2.2 Local Interpretation and Portfolio Tilts

Table 3 reports the results. Panel A shows that nearby managers tilt their portfolios toward greener stocks when local coverage interprets the disaster as climate change. A ten percentage point higher interpretation share raises their green tilt by 69.38 basis points at beginning-of-quarter prices and by 71.49 basis points at current prices in the quarter of the disaster (Columns 2 and 6). The estimates are similar with political controls (66.25 and 68.74 basis points, Columns 3 and 7) and with fund fixed effects (67.58 and 65.75, Columns 4 and 8).[34] In contrast, the volume of local coverage does not move the tilt, with coefficient on attention being small and never significant.

A one standard deviation increase in the interpretation share (16.4 percentage points) raises the tilt by 114 basis points, corresponding to 2 percent of the average tilt of nearby funds before the disaster. It also closes about a tenth of the gap in the tilt between a conventional fund and a fund with an ESG mandate, and it is 1.7 times the response to a local heat wave that causes injuries or deaths (Appendix Table E.19). In dollar terms, the average nearby fund, with $4.3 billion in assets, moves about $49 million from bottom-tercile to top-tercile stocks, and the median fund $7.7 million. Summed over the 29 nearby funds of the average disaster, which manage $125 billion, the shift reaches about $1.4 billion, or $110 million for the median disaster.

Figure 4 plots the effect quarter by quarter. The outcome is the change in the tilt since the end of the quarter before the disaster, in the specification of Table 3, Column 2. In the three quarters before the disaster, the estimates are close to zero and jointly not significant ($p = 0.36$ at beginning-of-quarter prices and $p = 0.33$ at current prices). The effect is 69.38 basis points in the quarter of the disaster, 52.57 in the next

[33] Political controls are the county's Democratic two-party vote share and the managers' political score, from the political donations of the named portfolio managers or, without them, of the management company. Appendix Table E.13 shows that the results are robust to including funds whose managers and management companies have no matched donations.

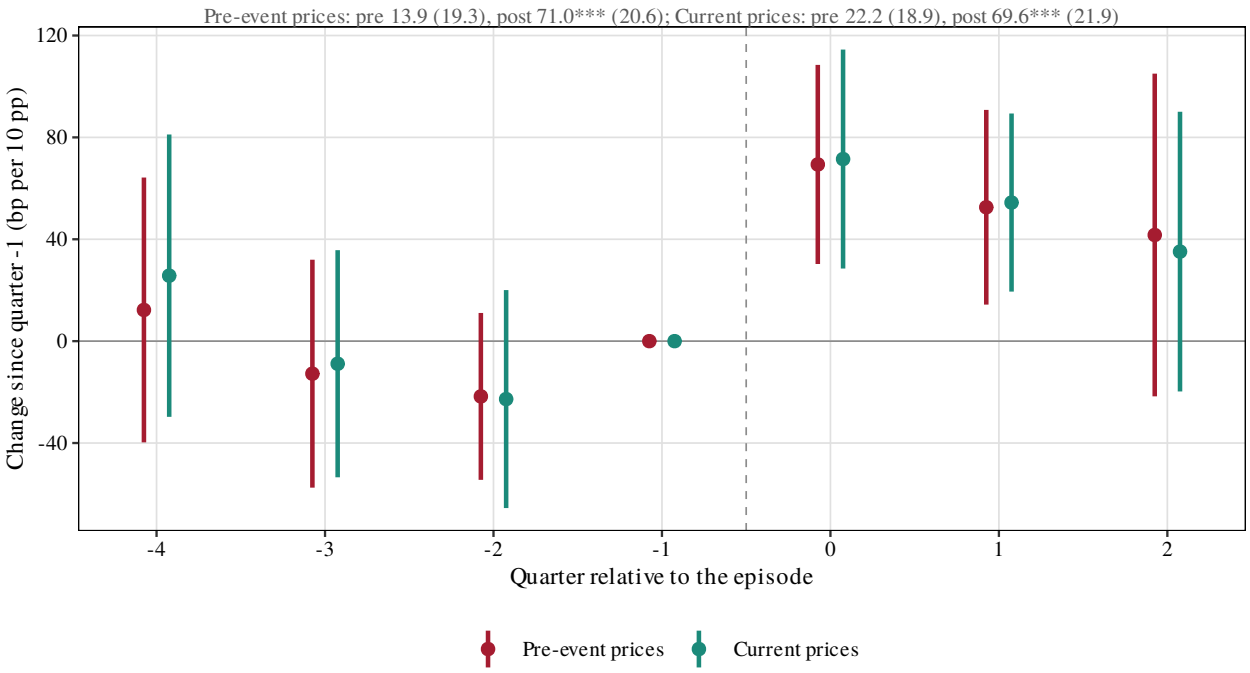
[34] The estimates are smaller but remain significant when I replace the band-by-disaster and county fixed effects with fixed effects for each combination of distance band, hazard and calendar quarter (Appendix Table E.13, Columns 5 and 10).

TABLE 3: MUTUAL FUND PORTFOLIO RESPONSE

	Dependent variable: response in basis points							
	Pre-event prices				Current prices			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A. Fund level: change in the fund's Environmental-Pillar tilt								
Interpretation	69.34***	69.38***	66.25***	67.58***	71.03***	71.49***	68.74***	65.75***
	(20.11)	(19.87)	(18.99)	(19.86)	(20.85)	(21.85)	(21.08)	(21.56)
Attention	−2.05	−.85	−1.09	−1.95	−4.52	−3.79	−4.01	−4.43
	(2.84)	(3.12)	(3.09)	(3.36)	(2.98)	(3.36)	(3.28)	(3.58)
Observations (fund-episodes)	37,047	37,047	37,047	36,964	37,047	37,047	37,047	36,964
Nearby fund-episodes	1,712	1,712	1,712	1,710	1,712	1,712	1,712	1,710
R^2	.031	.032	.033	.093	.044	.045	.045	.102
Exposure × episode FE	✓	✓	✓	✓	✓	✓	✓	✓
County FE	✓	✓	✓	✓	✓	✓	✓	✓
Fund style FE	✓	✓	✓	✓	✓	✓	✓	✓
Fund FE				✓				✓
Fund controls	✓	✓	✓	✓	✓	✓	✓	✓
Damage and coverage controls	✓	✓	✓	✓	✓	✓	✓	✓
Exposure × county controls		✓	✓	✓		✓	✓	✓
Exposure × political controls			✓	✓			✓	✓
Panel B. Stock level: change in the fund's weight in a stock, High-E minus Low-E								
Interpretation		3.57***	3.56***			3.66***	3.71***	
		(1.28)	(1.28)			(1.33)	(1.29)	
Attention		.04	.03			−.09	−.13	
		(.27)	(.28)			(.29)	(.30)	
Observations (fund-episode-stocks)		5,426,327	5,426,327			5,426,327	5,426,327	
Fund-episodes		53,028	53,028			53,028	53,028	
R^2		.577	.577			.586	.586	
Fund × episode FE		✓	✓			✓	✓	
Fund style × stock × episode FE		✓	✓			✓	✓	
Exposure × episode FE		✓	✓			✓	✓	
Damage and coverage controls		✓	✓			✓	✓	
Exposure × political controls			✓				✓	

Notes. This table reports estimates of equation (6) on the fund-episodes with the political measures. In Panel A the dependent variable is the change in the fund's Environmental-Pillar tilt over the episode quarter, winsorized at the 1st and 99th percentiles; in Panel B it is the change in the fund's weight in the stock, High-E minus Low-E. Rows report the effects among managers within 50 miles of the episode. Interpretation is per 10 percentage points and Attention per 10 log points. Each regression also includes Interpretation and Attention among affected managers and among managers more than 100 miles away; their coefficients are not reported. Fund controls are lagged log total net assets, log age and turnover; county controls are the county's interpretation of its other episodes, log deaths, log population, log population density, log income, college share, unemployment rate and urban share; political controls are the county Democratic two-party vote share and the managers' political score. Exposure is the distance band of the fund's offices: affected, within 100 miles and more than 100 miles away; county and political controls enter separately for each. Observations are weighted by the inverse number of episodes of the fund in the quarter. Standard errors, in parentheses, are clustered by episode. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

FIGURE 4: INTERPRETATION EFFECT AMONG NEARBY MANAGERS, BY QUARTER



Notes. Effect of local climate interpretation on the green tilt of nearby managers by quarter relative to the disaster, per 10 percentage points. The outcome is the change in the tilt since the end of quarter −1, in the specification of Table 3, Columns 2 and 6. Bars are 95 percent confidence intervals, with standard errors clustered at the disaster level. The estimates printed above the figure are from Appendix Table E.14, Columns 2 and 5.

quarter, and 41.68 two quarters later, when it is no longer significant. At current prices, the three estimates are 71.49, 54.42, and 35.15, and the last is not significant either. Pooling the quarters in one regression, as in Alok et al. [2020], gives 13.86 basis points, not significant, in quarters −4 and −3, and 71.03 basis points in quarters 0 to 2, relative to quarters −2 and −1 (Appendix Table E.14, Column 2).

The effect is concentrated near the disaster. Among managers more than 100 miles away, a ten percentage point higher interpretation share raises the tilt by 3.70 basis points, about a twentieth of the effect among nearby managers (Appendix Table E.18).

A disaster can move the fundamentals and prices of the firms it affects, and greener and browner firms need not be affected alike. A tilt measured at the level of the fund can then mix the managers' choices with changes in the stocks they hold. In Panel B, I estimate the regression for each fund, stock, and disaster, with the change in the fund's weight in the stock as the outcome. At this level, I can include fixed effects for each stock around each disaster, separately for each fund style. They account for everything that happens to the stock around the disaster and is common to the funds of that style that hold it, such as its return and fundamentals, the location of the firm and its own exposure to the disaster, national coverage of the firm and the shock to its industry. Fixed effects for each fund around each disaster absorb everything common to the fund's stocks. These include the attention and interpretation of its local coverage, its distance from the disaster, the damage in its county, its flows and its average trading, as well as the fund and county controls of Panel A. Interpretation, attention, damage and the band-by-disaster fixed effects therefore enter interacted with indicators for the top and bottom terciles of the Environmental Pillar score, as do the political controls in Columns 3 and 7. The Interpretation row reports the effect on top-tercile minus bottom-tercile stocks.

A ten percentage point higher interpretation share raises the weight of nearby managers in top-tercile relative to bottom-tercile stocks by 3.57 basis points at beginning-of-quarter prices and 3.66 at current prices, and the estimates are robust to the inclusion of political controls. This estimate compares how managers whose local coverage interprets the same disaster more or less as climate change reweight the same stocks, relative to the other stocks in their portfolios and to funds of the same style that hold the same stocks.

Alternative Explanations. The effect could capture fund flows and not the managers' own choices. Funds managed in the counties hit by the disaster can face outflows from local investors [Tubaldi, 2021], and these outflows raise two concerns. First, outflows can vary with interpretation for reasons unrelated to the managers' views. For instance, climate narratives may be less common in counties where the same disaster was more destructive, so that interpretation picks up differences in outflows. Second, the sales that outflows force follow their own pattern across stocks [Coval and Stafford, 2007], which can create a tilt even after adjusting for the rebalancing due to flows.

To address the first concern, I measure trades net of the proportional scaling of positions with flows. A ten percentage point higher interpretation share raises the net active purchases of nearby managers in top-tercile relative to bottom-tercile stocks by 30.14 basis points of lagged assets (Appendix Table E.15, Column 2), suggesting that tilt comes from the managers' own trade. To address the second, I estimate the effect on managers outside the counties hit, and I keep the affected managers in a band of their own. Nearby managers do not face the outflows of local investors in the counties hit, so forced sales cannot explain their tilt. Appendix Table E.18 reports the results for affected managers.

Alternatively, the effect could capture the sale of firms located in the disaster area, if managers expect these firms to perform worse or want to limit their investors' exposure, and these firms happen to be browner. To test this, I split the tilt of Table 3, Columns 2 and 6, into the weight moved across distance bands of the firms and the choice among firms at the same distance from the disaster (Appendix Table E.16). The component across bands is small and not significant, while the component within bands is almost identical to the main estimates. The effect therefore comes entirely from the choice among firms at the same distance from the disaster. At the stock level, the effect for firms outside the disaster area is the same as the baseline estimate (Appendix Table E.17). Within the disaster area, managers shift toward the greener firms even more, by 18.61 basis points. Managers thus do not sell local firms as a block, but reweight toward the greener ones.

Finally, the effect could reflect the funds' investors and not their managers. Works on mutual funds and local disasters typically place a fund at its headquarters and treat that address as the location of its managers [e.g., Fich and Xu, 2025]. The headquarters is also a natural proxy for where the fund's investors live, because individual investors favor locally managed funds [Bailey et al., 2011]. In my data, the two locations often differ. Among the 76,652 quarterly observations of funds with a managing office in Forms N-SAR and N-CEN or in Morningstar records of portfolio managers, the office is in the county of the headquarters in 65 percent of cases and more than 100 miles away in 30 percent. For sub-advised funds, the share more than 100 miles away is 63 percent.

Table E.21 uses this separation on the funds whose headquarters is known, with the specification of Table 3, Columns 1 and 2. In one regression, I estimate the effect for nearby managers whose headquarters is within 100 miles of the disaster, for nearby managers whose headquarters is farther away, and for funds whose headquarters is within 50 miles but whose managers are more than 100 miles away, using the interpretation of the headquarters county. The results show that nearby managers respond as much when their headquarters

is far as when it is near. Funds whose headquarters are near the disaster but whose managers are far respond less, but not significantly, suggesting that response depends more on manager location than on headquarters location.

Taken together, this evidence points towards the tilt being more likely driven by the managers' response to the local interpretation of the disaster compared to explanations involving fund flows, sale of firms in the disaster area, or pressures by the funds' investors.

4 Mechanisms

4.1 Salience

The pattern observed in green portfolio reallocation and fund flows and in mutual fund manager active rebalancing is consistent with salience and experiential updating. Both household advisers and mutual fund managers respond more strongly closer to the disaster (Tables E.10, 3, and E.18), while the response is more persistent among household advisers, remaining positive through six quarters (Table E.9). These patterns are consistent with documented salience effects of disasters. Alok et al. [2020] find that nearby mutual fund managers underweight disaster-zone stocks more than distant managers, with this aversion declining over time and distance. Gallagher [2014] finds that flood insurance take-up spikes after a flood and then declines to baseline, while take-up in nonflooded communities in the same television market rises at one third the rate. If salience is important, these differences in responses can matter for aggregation. Even under national coverage, investors closer to a disaster may react more strongly to the same news. This can change who the marginal buyer is and how investors who would usually accommodate the flow behave, because those investors may also increase their own demand for green assets.

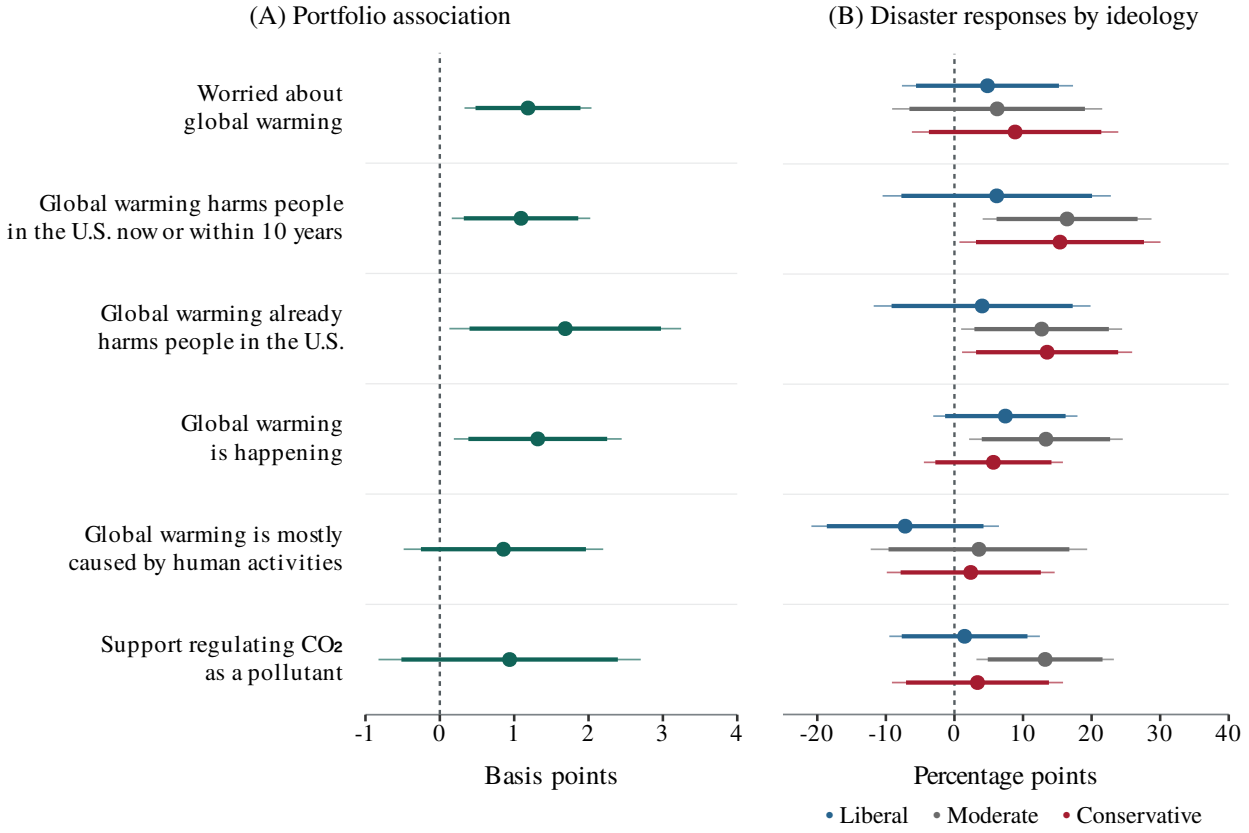
4.2 Taste or Belief shock

Changes in portfolio allocations can reveal preferences or beliefs shocks. In the previous sections, I discussed whether and whose beliefs or preferences are revealed by holdings in sustainable assets. Here, I focus on households and provide suggestive evidence on the mechanism through which extreme weather events and their interpretation affect portfolios. Household portfolios tend to be inert and therefore reflect beliefs only weakly [Giglio et al., 2021]. Since I found evidence of sizeable changes in portfolios, I now address what might trigger these changes and whether the response resembles a shock to tastes (attitudes) or a shock to climate concern.

Survey evidence supports a belief-based interpretation centered on perceived climate harm, with weaker evidence for beliefs about human causes and climate-policy attitudes. Figure 5A shows that, comparing media markets within the same year, advisers hold larger green ETF shares where more residents expect global warming to harm Americans already or within ten years. Associations with human-causation beliefs and regulatory support are positive but imprecisely estimated.

Perceived harm also responds to disaster experience. Following floods, moderates and conservatives become more likely to report harm occurring now or within ten years. Among conservatives, estimated changes in human-causation beliefs and regulatory support are much smaller and statistically insignificant. This contrast provides suggestive evidence that the perceived immediacy of climate consequences connects disaster experience to green investment, while changes in causal beliefs and policy attitudes receive less consistent support.

FIGURE 5: CLIMATE BELIEFS, GREEN INVESTMENT, AND DISASTER RESPONSES



Notes. Data are from Climate Change in the American Mind, 2009–2019. Panel A relates annual DMA belief shares to the green share of advisers' ETF holdings, with year fixed effects. Advisers are allocated to a DMA if they predominantly serve households and all their known U.S. offices lie within that DMA. Coefficients are basis points per 10-percentage-point higher belief share. Panel B reports post-disaster changes by ideology, in percentage points, using interviews in affected DMAs from 28 days before through 27 days after flood onset. Regressions include ideology indicators, DMA × survey-year, interview year-month and event fixed effects, and demographic and question-wording controls.

Binary outcomes equal one for the following answers: *Worried:* "How worried are you about global warming?" — somewhat or very worried. *Harm timing:* "When do you think global warming will start to harm people in the United States?" — "In 10 years" or "They are being harmed now"; the second indicator counts only the latter. *Warming happening:* "Do you think that global warming is happening?" and "How sure are you that global warming is happening?" — yes, with somewhat, very or extremely sure. *Human causation:* "Assuming global warming is happening, do you think it is. . . " — caused mostly by human activities. *Regulation:* "How much do you support or oppose the following policies?" — somewhat or strongly support "Regulate carbon dioxide (the primary greenhouse gas) as a pollutant."

Other valid answers equal zero. Thick and thin lines show 90% and 95% confidence intervals, respectively; standard errors are clustered by DMA.

Flood insurance. While the survey evidence provides a suggestive correlation between green investment and perceived climate harm, and shows the effect of disasters on the latter, it does not establish that these disaster-induced beliefs actually translate into financial behavior. First-time voluntary flood-insurance purchases provide more concrete evidence that extreme weather events affect households' financial decisions through the climate risk-perception channel, since insurance decisions are not governed by sustainability preferences.

I compare these purchases in affected counties with those in unaffected counties of the same states, separately after floods and wildfires.[35] This comparison helps distinguish interpretation from salience. Floods make flood risk salient directly; wildfires allow me to examine whether climate attribution is associated with demand for protection against another hazard.

By month three, the estimated increase relative to unaffected counties is 41 versus 15 policies per 100,000 residents at the 75th and 25th percentiles of climate attribution after wildfires, compared with 43 versus 40 after floods. The flood response is consistent with salience and experiential learning [Gallagher, 2014], while the cross-hazard insurance response to wildfires suggests that climate interpretation matters for households' demand for protection beyond disaster salience.

4.3 Return Expectations

Return expectations and national news. Affected advisers may buy green ETFs because they expect higher returns [Giglio et al., 2023]. Since a single local disaster event provides no information about either climate risk trends or green firms' cash flows, the return expectation motive requires anticipating that many investors will buy these products as a result of the event[36] and push up their prices. These expectations would be shared by affected and unaffected advisers alike, regardless of their location. The common component of this response is therefore absorbed by the episode fixed effects. For return expectations to explain the stronger response of affected and nearby advisers, and the interpretation gradient among them, affected advisers in higher-interpretation places would need to expect a stronger reaction from others. In particular, they would need to expect a stronger reaction than unaffected advisers do, and than affected advisers in places where the disaster was interpreted less as a climate event. I therefore test whether the response to local interpretation remains after accounting for national attention and national interpretation, and whether either measure changes its strength. I add national coverage and national framing, alone and interacted with local framing, to the county and adviser regressions of Section 3 (Table OA.C.1). The local interpretation response among affected counties and advisers remains positive after accounting for national news. If anything, national attention strengthens this response,[37] while national framing does not significantly moderate this response. National attention widens the response gap between affected places with high and low local interpretation, consistent with polarization of allocation responses.

Finally, if advisers instead bought green ETFs to chase past returns, the response would be larger after green ETFs had done well. The response is the same after strong and weak years for green ETFs (Table G.2).

[35] I estimate local projections of cumulative first-time voluntary NFIP issuance per 100,000 residents from the disaster month through month h. County, episode and state-by-event-year fixed effects absorb persistent county differences and shared shocks and trends, so the identifying variation comes from how insurance purchases vary with climate attribution in affected relative to unaffected counties, conditional on total disaster and climate-only coverage. Standard errors are clustered by county.

[36] Morris and Shin [2002] show how public information can influence behavior through expectations of others' actions. Schmidt-Engelbertz and Vasudevan [2025] document speculation motivated by beliefs about other investors, even when it conflicts with investors' own valuations.

[37] The strengthening is statistically significant for counties and book-weighted advisers, and remains positive but is not statistically significant with equal adviser weights.

5 Identification

Instrument for interpretation. One concern is that the local climate interpretation of a disaster proxies for the county's pre-existing climate beliefs and green attitudes. To address this, I use supply-side shocks to media interpretation.

Larger newspaper chains are more distant from the counties they serve and supply centrally produced copy; the 2019 Gannett-Gatehouse merger is one instance of the ownership changes that move the instrument.

For newspaper o in calendar quarter t, I define sister_count_{ot} as the number of other newspapers in the sample held by the outlet's strictly prior owner. I estimate the outlet equation

$$M_{ot} = \pi \log(1 + \text{sister_count}_{ot}) + \rho \log(1 + A_{ot}) + \alpha_o + \tau_t + u_{ot}, \tag{7}$$

on 2015–2024 newspaper–quarters with $A_{ot} \geq 1$ whose prior owner held more than one other newspaper in the sample, where α_o are newspaper fixed effects and τ_t calendar-quarter fixed effects, with standard errors clustered by newspaper. The county instrument is constructed as the circulation-weighted mean of chain size across the chained newspapers serving county c,

$$Z_{c,t(e)} = \frac{\sum_{o \in C_{c,t(e)}} w^D_{oc} \log\big(1 + \text{sister_count}_{o,t(e)}\big)}{\sum_{o \in C_{c,t(e)}} w^D_{oc}}, \tag{8}$$

where C_{ct} is the set of chained newspapers serving county c under strictly prior ownership and w^D_{oc} is the fixed circulation weight.

The instrument varies at sell-offs, purchases and mergers of the county's outlets, decisions taken at the chain level for reasons unrelated to the climate attitudes of the local newsroom and of the county's readers.

Seven parents hold 382 of the 672 major US dailies.[38] Outside the United States, newspapers held by one owner share that owner's slant [Garz and Rickardsson, 2023], and acquisitions shift slant in a direction that depends on the acquirer [Ananyev and Volkova, 2024]; in American local broadcasting, ownership changes shift content [Martin and McCrain, 2019]. Acquisitions and consolidation in the industry follow declining advertising revenue, the liquidity needs of legacy owners, and cost cutting in roll-ups [Ewens et al., 2022, Gao et al., 2020].

The climate-interpreted share of disaster coverage differs persistently across owners, and newspaper chain size predicts that share at the outlet level; under a larger prior owner, local disaster coverage has a lower climate-interpreted share (Table F.1). The outlet interpretation share M_{ot} is stable within an owner and shifts at an ownership change.

Instrument for attention. A second concern is that the interpretation measure captures the salience of the disaster itself. Disasters that attract greater attention may also lead newspapers to discuss their climate implications more extensively. The estimated interpretation effect could therefore reflect how much attention the event receives.

I address this concern in two steps. First, I instrument climate interpretation using newspaper-chain ownership while controlling for the volume of disaster coverage. This specification treats coverage as

[38] The News Media Alliance counts 672 major US dailies, of which 382 are held by seven parents: Gannett (250), Lee Enterprises (90), Alden Global Capital / Digital First Media (56), Chatham Asset Management / McClatchy (30), Hearst (23), Advance Local (22) and Tribune (10). Harvard Future of Media Project, *Index of Seven Big Owners of Dailies*, static index time-stamped 11 May 2021.

exogenous conditional on the controls and fixed effects. It asks whether ownership-induced differences in interpretation predict portfolio responses at a given level of observed coverage. A causal interpretation requires the ownership instrument to affect portfolios through interpretation, without also shifting unmeasured attention that independently affects investment.

Separating attention from interpretation requires holding fixed the incentive of the county's newspapers to cover the event. Beyond disaster coverage itself, I control for newspaper-audience overlap with the event footprint outside the county, inspired by the newspaper-market congruence designs of Snyder and Strömberg [2010] and Myers [2026]. Instrumenting coverage jointly with this measure gives similar interpretation estimates from a weakly identified system; that version is in the Online Appendix.

Consider three nearby counties, A, B, and C. A disaster hits only A. People in B mostly read newspaper X, which has 90 percent of its readers in A. People in C mostly read newspaper Y, which has no readers in A. X therefore has more incentive than Y to cover the disaster, so people in B should receive more coverage than people in C, even though neither county is hit. The intended variation captures the event's newsworthiness to the newspapers serving a county, conditional on the included disaster characteristics.

For each newspaper, I sum its fixed audience shares across the counties in the disaster footprint, excluding the county whose investment response is being studied. Each share retains the newspaper's total audience as its denominator. I then average these shares across newspapers serving the county, using fixed circulation weights w^D_{oc}, normalized by their sum. For a county outside the footprint, this includes readership throughout the affected area; for a directly affected county, it includes readership only in the other affected counties. The instrument therefore applies to both directly affected and nearby counties.

At the outlet level, chain size moves the interpretation share and leaves disaster coverage unmoved (Table F.1).

I estimate:

$$Y_{ue} = \beta_M M_{ce} + \beta_A A_{ce} + X'_{ce}\gamma + \mu_c + \tau_{t(e)} + \lambda_{h(e)\times b(c,e)} + \varepsilon_{ue}, \tag{9}$$

where u is a county or an adviser in county c, A_{ce} is local disaster coverage in logs, X_{ce} collects the logs of one plus own-county deaths and damage and, in columns 2 and 4 of Table 4, the county Democratic two-party vote share at the previous presidential election and the audience overlap of the county's newspapers with the event footprint, μ_c are county fixed effects, $\tau_{t(e)}$ calendar-quarter fixed effects, and $\lambda_{h\times b}$ distance-band by hazard-family fixed effects; the comparison is among counties at the same exposure to disasters of one hazard family. M_{ce} is instrumented with the size of the newspaper chain owning the county's outlets. Adviser regressions weight each adviser by its share of the county-episode ETF book in the quarter before the episode. Standard errors are clustered by county.

Table 4 reports the estimates. Panel A uses the county response of Table 2, Panel B the adviser response of Table E.11; all columns instrument interpretation with chain size, and columns 2 and 4 add the political and audience-overlap controls. Per 10 percentage points of the interpretation share, the instrumented response is 22 to 25 basis points for counties and 19 to 22 basis points for advisers, in line with the local-projection estimate of Table 2 (19 basis points per 10 points); the weak-instrument-robust confidence sets exclude zero at the 10 percent level.

6 Alternative Explanations

I address here the possible alternative explanations that would threaten either the reading of my previous results as the effect of the interpretation of the specific event, or the interpretation of the observed response

TABLE 4: GREEN-ETF PORTFOLIO RESPONSE TO INSTRUMENTED INTERPRETATION

	Dependent variable:			
	Green-ETF share response (basis points)			
	Pre-event prices		Current prices	
	(1)	(2)	(3)	(4)
Panel A. Counties				
Interpretation	22.45	22.47	24.59*	24.60*
	(14.20)	(14.21)	(14.62)	(14.59)
First stage: chain size	−8.90***	−8.82***	−8.90***	−8.82***
	(2.98)	(2.89)	(2.98)	(2.89)
Reduced form: chain size	−19.99*	−19.83*	−21.89*	−21.71*
	(10.73)	(10.74)	(11.18)	(11.20)
Observations	694	694	694	694
First-stage F	8.91	9.33	8.91	9.33
Anderson–Rubin p	.064	.067	.052	.054
AR 90% set	[2.5, 62.2]	[2.3, 61.2]	[3.8, 65.2]	[3.6, 63.9]
Panel B. Advisers				
Interpretation	19.28*	19.18*	22.48**	22.38**
	(10.49)	(10.53)	(11.31)	(11.31)
First stage: chain size	−8.53***	−8.51***	−8.53***	−8.51***
	(2.82)	(2.77)	(2.82)	(2.77)
Reduced form: chain size	−16.45**	−16.32**	−19.17**	−19.04**
	(7.50)	(7.55)	(8.51)	(8.54)
Observations	4,342	4,342	4,342	4,342
County-events	660	660	660	660
First-stage F	9.17	9.44	9.17	9.44
Anderson–Rubin p	.030	.032	.025	.027
AR 90% set	[4.7, 48.6]	[4.5, 48.2]	[6.3, 53.2]	[6.0, 52.5]
First-stage F, outlet level	20.85	20.85	20.85	20.85
Coverage; own-county deaths and damage	✓	✓	✓	✓
Democratic vote share; audience reach		✓		✓
County, quarter and hazard × distance-band FE	✓	✓	✓	✓

Notes. 2SLS estimates using circulation-weighted log newspaper-chain size as the instrument. Outcomes follow Tables 2 (counties) and E.11 (advisers); advisers are weighted by their pre-event county-episode ETF-book share. Interpretation effects are basis points per 10-percentage-point increase. First-stage and reduced-form effects are per standard deviation of chain size, in percentage points and basis points, respectively. Anderson–Rubin inference is robust to weak instruments. Standard errors are clustered by county. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

as evidence of revealed sustainability demand shock.

6.1 Threats to Interpretation

Interpretation or Generic Climate Salience.

One possibility is that interpretation effect is simply capturing general contemporaneous or diffused climate talk or attention unrelated to the event. Table F.4 addresses that concern directly by adding my corresponding local measures of same-quarter and lagged non-disaster-specific climate attention to the analysis. I included these terms both as controls and as heterogeneities, allowing generic non-disaster climate salience (e.g., due to climate policy events occurring in the same quarter) to compete against disaster interpretation in mediating the portfolio response. The climate talk interactions are all indistinguishable from zero, while the interpretation coefficient, estimated against these competing moderators, remains positive at 29.0bps in the most demanding case (against a baseline result of 30.31bps).

Interpretation of Current Event or Predetermined County Type or Predisposition. While I ruled out that a higher volume of disaster coverage is driving the results, a remaining concern is that the coefficient on climate change contextualization may be driven by counties where climate is already more salient, more discussed, or more behaviorally relevant and, therefore, may be picking up pre-existing local concern about climate change and not how disaster experiences are interpreted.

One possible explanation of the response is that it is driven by attitudes the county already held, with the disaster as the occasion on which they are acted upon. If residents more concerned about climate change buy green assets whenever a salient local event occurs, and if coverage in those same counties is also more climate-interpreted, this would imply a spurious correlation between interpretation and green reallocation. Table F.4 suggests that this is not the case. Each row adds one county characteristic, interacted with the disaster-zone indicator, with climate interpretation, and with their product; the four rows here use the county-level share of individuals worried about global warming, the perceived scientific consensus, the county Democratic vote share, and the share of population that has college degree. All four triple interactions are statistically indistinguishable from zero, and the interpretation effect evaluated at the mean of each remains between 36.7 and 45.2 basis points, significant at the 1% level. Measured climate attitudes have little explanatory power for the magnitude of the interpretation response.

If local climate interpretation mainly captures a general propensity of climate-concerned counties to buy green assets after salient local shocks, the same pattern should appear after earthquakes. I estimate the response of green ETFs to earthquakes, while assigning to each county its local climate interpretation of other disasters, measured in the event quarter or averaged through the previous quarter, capturing its local predisposition for interpreting events as climate related. There is no statistically significant differential green tilt after earthquakes in counties with more concurrent climate interpretation (Table F.3).

6.2 Threats to Event Link and Specificity

Comparable or Same Event. To compare portfolio responses to the same disaster, I use advisers in affected counties and nearby advisers within 50 miles of the affected area, measuring local interpretation at the adviser's office ZIP. I estimate

$$\Delta_h S_{ie} = \theta_h M_{ie} + \alpha^h_{e,s(i)} + \lambda^h_{e,b(i,e)} + \mu^h_{c(i)} + X'_{ie}\gamma_h + u_{ie,h}, \qquad h \in \{0, 1\}, \tag{10}$$

where $\Delta_h S_{ie}$ is the change in green-ETF share from the preceding quarter to the disaster quarter or the following quarter, measured in basis points at pre-event prices. M_{ie} measures interpretation in ten-percentage-point units, with a common coefficient across affected and nearby advisers. X_{ie} includes the controls and their group interactions described in the Online Appendix.

I compare portfolio changes among advisers facing the same disaster in the same state whose local coverage interprets it differently. Episode×band fixed effects allow affected advisers and nearby advisers 0–25 and 25–50 miles from the affected area to have different average responses to each disaster. With county fixed effects, the comparison is relative to the county's average portfolio response across episodes in which its advisers are affected or nearby.

A ten-percentage-point increase in interpretation is associated with a 7-basis-point increase in green-ETF share in the disaster quarter and a 12-basis-point increase by the following quarter, both relative to the preceding quarter (Table F.2, column 3). Standard errors are clustered by disaster episode and ZIP.

6.3 Threats to Revealed Demand Shock

Household Demand Shocks or Advisers. I restrict the sample to local advisers serving predominantly household accounts. Still, most assets are managed on a discretionary basis, so portfolio changes can reflect both client preferences and advisers' own views. I therefore compare county voting, which proxies for clients' political views, with advisers' political views measured from staff donations, including both in the regression. If advisers' own views drive the response, their political leanings should help explain it. The response is stronger in less Democratic counties but does not vary significantly with advisers' own political views (Table G.1). This supports a household-demand interpretation, consistent with Pan et al. [2025], who find that county political preferences predict household-serving advisers' portfolio choices even after accounting for advisers' own political leanings.

The effect is largest where the two parties are near even, and it is also positive and significant in Republican-majority counties, while it is not statistically distinguishable from zero in strongly Democratic counties. This pattern is consistent with the survey evidence, although the differences across political groups are not jointly significant (Appendix Table G.1).

Less Democratic advisers are less likely to hold any green ETF to begin with. However, conditional on holding a green ETF, their own views do not significantly affect the disaster response, and, more importantly, the interpretation effect[39]. Taken together, this evidence suggests that the green ETF response to the local interpretation records the beliefs of the households the advisers serve, and not the advisers' own.

Within-ETF Reallocation or Liquidity Shocks. One concern might be that the effect could be purely mechanical: disasters may generate liquidity needs, withdrawals, or precautionary adjustments that change ETF shares even without any active move toward green assets. In particular, advisers may cut ETFs first to meet their liquidity needs because ETFs tend to be particularly liquid and easy to trade, and may be held as a liquidity buffer. If among ETFs, non-green ETFs are liquidated even more aggressively, then the ETF denominator can fall and green ETF shares can rise even without a genuine tilt toward green funds. Instead, I find that green ETF assets rise alongside non-green and total ETF assets, while total observed 13F

[39]Only smaller advisers show an effect of the firm's own leaning on the average disaster response. Among the green ETF holders, more Republican staff means a smaller adjustment after the disaster, by 12 basis points per standard deviation of the firm's lean. The effect sits among the firms with small ETF books, the bottom third, with a median book of eleven million dollars, about two percent of their regulatory assets, and is absent when advisers are weighted by their share of the county's book. They still show none on the interpretation response.

holdings show no statistically significant change (Table OA.E.1). ETF and whole-portfolio flows show no systematic contraction in counties and episodes with higher climate interpretation (Table OA.E.2). Moreover, the interpretation effect holds also among advisers in nearby counties not directly affected by the extreme weather event. These findings do not support liquidation as the explanation for the increase in the green ETF share.

Conflict of Interest. A separate concern is that advisers encourage clients to buy their green products, or to buy green ETFs and then accommodate that demand by selling the underlying stocks. About 2% of the local advisers in my sample sponsor or manage the ETFs in their holdings. I find that when advisers purchase green ETFs, they also tend to purchase the underlying constituent stocks directly (Table OA.E.3). This pattern does not support systematic offsetting sales within the observed portfolios. Advisers may also use a disaster as an occasion to rebalance client portfolios and justify their fees. I control for advisers' prior ETF and whole-portfolio turnover. The interpretation response does not vary significantly with a one-standard-deviation increase in either measure (Table OA.E.4; 2015–2024 estimates in Table OA.E.5).

7 Aggregate Implications of Climate Interpretation

At the aggregate level, existing evidence links disasters to increased flows into green ETFs and to the relative outperformance of sustainable assets [Fiordelisi et al., 2023, Misev and Balles, 2024].

To complement this evidence, I compare how flows into ESG-labelled funds relative to other funds respond to national disaster-attention surprises at different levels of national climate interpretation. Attention is measured through national newspaper coverage, and interpretation is the share of disaster articles linking events to climate change. The analysis uses my sample of US domestic-equity mutual funds and ETFs over 2015–2024.[40]

At the average national interpretation share of 25.6%, a one-standard-deviation national disaster-attention surprise is associated with an estimated cumulative increase in flows into ESG-labelled funds relative to other funds equivalent to approximately $288 million over the attention month and the following two months. Figure 6 compares the responses to the same attention surprise at the 25th and 75th percentiles of the national interpretation share, corresponding to 17.9% and 33.9% of coverage. The estimated three-month response is approximately $580 million larger at the 75th than at the 25th percentile.[41]

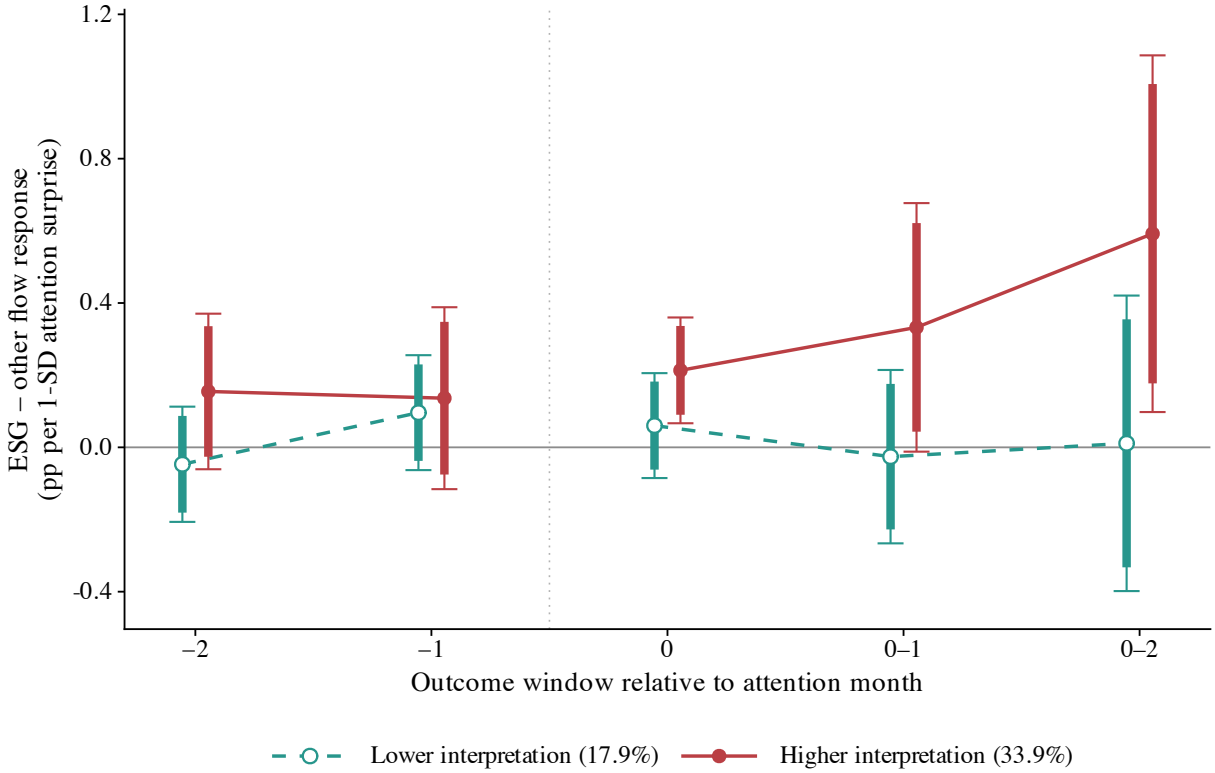
Van der Beck [2026], in turn, shows that ESG flows and, more generally, institutional ESG demand are important drivers of green-minus-nongreen stock returns. Together, these findings motivate asking how much additional aggregate ESG demand and relative appreciation could arise when investors interpret salient disasters more strongly as manifestations of climate change. I examine this question through a counterfactual that applies the locally estimated interpretation response to a broader investor population and uses the calibrated market response to quantify the resulting price adjustment (Appendix H).

I consider an increase in local climate framing from its 25th to its 75th percentile in my sample, holding newspaper coverage at its median and the remaining controls at their sample averages. The estimated

[40] The specification, adapted from Hartzmark and Sussman [2019], compares ESG-labelled funds with other funds within the same investment category and calendar month through investment-category-by-month fixed effects. It controls for fund characteristics and weights observations by log lagged AUM.

[41] In a back-of-the-envelope calculation following Hartzmark and Sussman [2019], I multiply each month's estimated relative flow response by each ESG fund's preceding-month assets and sum across funds and the three response months. The reported amounts average these three-month totals across the 118 eligible starting dates. The response at average interpretation is imprecisely estimated: its 90% and 95% confidence intervals include zero.

FIGURE 6: NATIONAL ATTENTION AND FUND FLOWS: LOG-AUM WEIGHTS



Notes. This figure plots the adjusted flow-rate responses of ESG-labelled funds relative to other funds to the same positive one-standard-deviation national disaster-attention surprise, evaluated at the 25th and 75th percentiles of national interpretation (17.9% and 33.9%). The sample pools US domestic-equity mutual funds and ETFs over 2015–2024, with log lagged-AUM regression weights. Monthly percentage flows are return-adjusted net flows divided by preceding-month assets. Outcomes at −2, −1 and 0 are monthly rates; 0–1 and 0–2 sum monthly rates over two and three months. The specifications include investment-category-by-month fixed effects and lagged fund size, age, returns and expenses. Thin and thick bars show 95% and 90% confidence intervals, based on monthly-score Newey–West covariance estimates with 12 lags. The −2 and −1 points describe associations with past flows conditional on origin-time controls. Samples vary across outcome windows. Figure H.2 reports the equal-weight specification.

additional ESG allocation is 31.65 basis points over the event quarter and the following quarter. To interpret this response as a demand shift at initial prices, I assume that, conditional on the specification, the comparison groups have the same average responsiveness of ESG allocations to relative fund prices, so that their responses to the common price movement cancel from the comparison.

The evidence presented in Section 3.1 that relative flows into ESG-labelled mutual funds also respond to interpretation supports extending the calculation from ETFs to mutual funds and ETFs together. The evidence on rebalancing towards greener stocks by mutual-fund managers without ESG mandates motivates a further extension to institutional investors' equity demand. The central assumption is that investors activated in my counterfactual analyses would respond to stronger climate interpretation as the disaster-exposed investors in my sample do, at the specified level of attention and information environment.

Assume additional ESG purchases are financed by reducing other holdings. Accounting for the ESG holdings already contained in the broad market portfolio, the fund-sector calculation implies an additional ESG-versus-benchmark reallocation of approximately 0.320% of capital. I apply this rate to the $24.74 trillion of institutional equity holdings covered by the calibration, counting the underlying capital once.

Purchasing the ESG portfolio and selling the broad market portfolio generates net purchases of stocks overweighted by ESG funds and net sales of underweighted stocks. I translate this demand into valuations using van der Beck's [2026] ESG flow multiplier, $\kappa_{\text{ESG}} \simeq 0.72$, which measures the difference in green and nongreen stocks' dollar valuation changes per dollar reallocated. Expressing the same response in return units gives $M_{\text{ESG}} \simeq 1.12$, relating the green-minus-nongreen return to relative demand normalized by each group's market capitalization.[42] This way I can incorporate into my counterfactual the investors' willingness to accommodate trades as prices change, under the assumption of comparable portfolio demand and market accommodation.

To compare with annual returns, consider disasters occurring throughout a year that collectively activate the institutional investor base,[43] with each investor completing the estimated additional adjustment once during the year. I compare their reallocations under stronger and weaker climate interpretation, holding attention and national news fixed. Under these assumptions, stronger interpretation generates $79.21 billion of additional ESG-versus-benchmark demand, implying $56.81 billion in additional green-minus-nongreen valuation change and 33.92 basis points of additional return differential. This additional appreciation is equivalent in magnitude to 18.2% of the average annual return contribution attributed to institutional ESG demand and 15.2% of the observed annual return spread.

8 Conclusion

[42] Both multipliers are calibrated using van der Beck's [2026] December 2022 inputs.

[43] See Appendix H for the same exercise applied to demand for (i) green ETFs alone and (ii) ESG-labelled mutual funds and ETFs together.

References

John M. Abowd, Francis Kramarz, and David N. Margolis. High wage workers and high wage firms. *Econometrica*, 1999.

Georgij Alekseev, Stefano Giglio, Quinn Maingi, Julia Selgrad, and Johannes Stroebel. A quantity-based approach to constructing climate risk hedge portfolios. Working paper, October 2, 2024 version, 2024. URL https://stefanogiglio.org/papers/alekseev-giglio-maingi-selgrad-stroebel-2024.pdf.

Georgij Alekseev, Stefano Giglio, Quinn Maingi, Julia Selgrad, and Johannes Stroebel. A quantity-based approach to constructing climate risk hedge portfolios. *Journal of Financial Economics*, forthcoming.

Shashwat Alok, Nitin Kumar, and Russ Wermers. Do fund managers misestimate climatic disaster risk? *Review of Financial Studies*, 33(3):1146–1183, 2020. doi: 10.1093/rfs/hhz143.

Maxim Ananyev and Ekaterina Volkova. Media ownership and ideological slant: Evidence from Australian newspaper mergers. *PLOS ONE*, 19(12):e0315137, 2024. doi: 10.1371/journal.pone.0315137.

Anders Anderson and David T. Robinson. Populism, polarization and green investment. Working Paper 32131, National Bureau of Economic Research, 2026. URL https://www.nber.org/papers/w32131. May 2026 revision; originally issued February 2024.

Ian R. Appel, Todd A. Gormley, and Donald B. Keim. Passive investors, not passive owners. *Journal of Financial Economics*, 121(1):111–141, 2016.

David Ardia, Keven Bluteau, Kris Boudt, and Koen Inghelbrecht. Climate change concerns and the performance of green vs. brown stocks. *Management Science*, 2023. doi: 10.1287/mnsc.2022.4636.

Elliott Ash, Anton Boltachka, Sergio Galletta, and Matteo Pinna. Media bias and climate change skepticism. Working Paper 05/2023, ETH Zurich, Center for Law & Economics, 2023. Version of November 14, 2023.

Susan Athey, Markus Mobius, and Jeno Pal. The impact of aggregators on internet news consumption. Working Paper 28746, National Bureau of Economic Research, 2021.

Bok Baik, Jun-Koo Kang, and Jin-Mo Kim. Local institutional investors, information asymmetries, and equity returns. *Journal of Financial Economics*, 97(1):81–106, 2010.

Warren Bailey, Alok Kumar, and David Ng. Behavioral biases of mutual fund investors. *Journal of Financial Economics*, 102(1):1–27, 2011. doi: 10.1016/j.jfineco.2011.05.002.

Malcolm Baker, Mark L. Egan, and Suproteem K. Sarkar. Demand for ESG. Working Paper 30708, National Bureau of Economic Research, 2024. URL https://www.nber.org/papers/w30708. Originally issued December 2022; revised December 2024.

Ray Ball and Lakshmanan Shivakumar. How much new information is there in earnings? *Journal of Accounting Research*, 46(5):975–1016, 2008. doi: 10.1111/j.1475-679X.2008.00299.x.

Gennaro Bernile, Alok Kumar, and Johan Sulaeman. Home away from home: Geography of information and local investors. *The Review of Financial Studies*, 28(7):2009–2049, 2015. doi: 10.1093/rfs/hhv004.

Board of Governors of the Federal Reserve System. Changes in U.S. family finances from 2019 to 2022: Evidence from the survey of consumer finances. Technical report, Federal Reserve Board, 2023. Federal Reserve Bulletin; Box 2, direct and indirect holdings of publicly traded stock.

Joan Calzada and Ricard Gil. What do news aggregators do? evidence from google news in spain and germany. *Marketing Science*, 2020.

Darwin Choi, Zhenyu Gao, and Wenxi Jiang. Attention to global warming. *Review of Financial Studies*, 33 (3):1112–1145, 2020.

J. Anthony Cookson, Joseph E. Engelberg, and William Mullins. Does partisanship shape investor beliefs? Evidence from the COVID-19 pandemic. *Review of Asset Pricing Studies*, 10(4):863–893, 2020. doi: 10.1093/rapstu/raaa018.

Joshua Coval and Erik Stafford. Asset fire sales (and purchases) in equity markets. *Journal of Financial Economics*, 86(2):479–512, 2007.

Tatyana Deryugina. How do people update? the effects of local weather fluctuations on beliefs about global warming. *Climatic Change*, 118(2):397–416, 2013. doi: 10.1007/s10584-012-0615-1.

Milena Djourelova, Ruben Durante, Elliot Motte, and Eleonora Patacchini. Experience, narratives, and climate change beliefs. Discussion Paper DP18738, CEPR, 2024.

Arindrajit Dube, Daniele Girardi, Òscar Jordà, and Alan M. Taylor. A local projections approach to difference-in-differences. *Journal of Applied Econometrics*, 2025. doi: 10.1002/jae.70000. Also NBER Working Paper 31184, April 2023, revised May 2025.

Joseph E. Engelberg and Christopher A. Parsons. The causal impact of media in financial markets. *Journal of Finance*, 66(1):67–97, 2011.

Michael Ewens, Arpit Gupta, and Sabrina T. Howell. Local journalism under private equity ownership. Working Paper 29743, National Bureau of Economic Research, February 2022.

Sean Ewing. Profit over public good: The impact of investment firm ownership on local news and political behavior. *Political Communication*, 43(3):501–538, 2026. doi: 10.1080/10584609.2025.2601736.

Eliezer M. Fich and Guosong Xu. Do salient climatic risks affect shareholder voting? *Review of Finance*, 29(2):567–602, 2025.

Franco Fiordelisi, Giuseppe Galloppo, and Viktoriia Paimanova. Climate change shocks and socially responsible investments. *Business Ethics, the Environment & Responsibility*, 32(1):40–56, 2023. doi: 10.1111/beer.12477. URL https://doi.org/10.1111/beer.12477.

Justin Gallagher. Learning about an infrequent event: Evidence from flood insurance take-up in the united states. *American Economic Journal: Applied Economics*, 6(3):206–233, 2014. doi: 10.1257/app.6.3.206.

Pengjie Gao, Chang Lee, and Dermot Murphy. Financing dies in darkness? the impact of newspaper closures on public finance. *Journal of Financial Economics*, 135(2):445–467, 2020.

Marcel Garz and Jonna Rickardsson. Ownership and media slant: Evidence from Swedish newspapers. *Kyklos*, 76(1):18–40, 2023. doi: 10.1111/kykl.12318.

Stefano Giglio, Matteo Maggiori, Johannes Stroebel, and Stephen Utkus. Five facts about beliefs and portfolios. *American Economic Review*, 111(5):1481–1522, 2021. doi: 10.1257/aer.20200243.

Stefano Giglio, Matteo Maggiori, Johannes Stroebel, Zhenhao Tan, Stephen Utkus, and Xiao Xu. Four facts about ESG beliefs and investor portfolios. Working Paper 31114, National Bureau of Economic Research, 2023.

Stefano Giglio, Matteo Maggiori, Johannes Stroebel, Zhenhao Tan, Stephen Utkus, and Xiao Xu. Four facts about ESG beliefs and investor portfolios. *Journal of Financial Economics*, 164:103984, 2025. doi: 10.1016/j.jfineco.2024.103984.

Samuel M. Hartzmark and Abigail B. Sussman. Do investors value sustainability? a natural experiment examining ranking and fund flows. *Journal of Finance*, 74(6):2789–2837, 2019. doi: 10.1111/jofi.12841.

Investment Company Institute. 2025 investment company fact book. Technical report, Investment Company Institute, 2025. URL https://www.ici.org/files/2025/2025-factbook.pdf. Chapter 4, US exchange-traded funds, Figure 4.7; household ownership figures are for 2024.

Marcin Kacperczyk, Clemens Sialm, and Lu Zheng. Unobserved actions of mutual funds. *Review of Financial Studies*, 21(6):2379–2416, 2008.

Eugene Kandel and Neil D. Pearson. Differential interpretation of public signals and trade in speculative markets. *Journal of Political Economy*, 103(4):831–872, 1995. doi: 10.1086/262005.

Byoung Uk Kang and Jonathan Sangwook Nam. Do local newspapers matter to institutional investors? *Contemporary Accounting Research*, 42(3):1713–1743, 2025. doi: 10.1111/1911-3846.13049.

Elisabeth Kempf, Mancy Luo, Larissa Schäfer, and Margarita Tsoutsoura. Political ideology and international capital allocation. *Journal of Financial Economics*, 148(2):150–173, 2023. doi: 10.1016/j.jfineco.2023.02.005.

Gary King, Benjamin Schneer, and Ariel White. How the news media activate public expression and influence national agendas. *Science*, 358(6364):776–780, 2017. doi: 10.1126/science.aao1100.

Patrick Kline, Raffaele Saggio, and Mikkel Sølvsten. Leave-out estimation of variance components. *Econometrica*, 2020.

Ralph S. J. Koijen and Motohiro Yogo. A demand system approach to asset pricing. *Journal of Political Economy*, 127(4):1475–1515, 2019. doi: 10.1086/701683.

Ralph S. J. Koijen, Robert J. Richmond, and Motohiro Yogo. Which investors matter for equity valuations and expected returns? *Review of Economic Studies*, 91(4):2387–2424, 2024.

Anthony Leiserowitz, John Kotcher, Seth Rosenthal, Emily Goddard, Jennifer Carman, Marija Verner, Teresa Myers, and Edward Maibach. Climate change in the american mind: Beliefs & attitudes, spring 2026. Yale Program on Climate Change Communication, 2026. URL https://climatecommunication.yale.edu/publications/climate-change-in-the-american-mind-beliefs-attitudes-spring-2026/.

LSEG. Environmental, social and governance scores from LSEG. Methodology, LSEG Data & Analytics, October 2024.

Ben R. Marshall, Hung T. Nguyen, Nhut H. Nguyen, Nuttawat Visaltanachoti, and Martin Young. Do climate risks matter for green investment? *Journal of International Financial Markets, Institutions and Money*, 75:101438, 2021. doi: 10.1016/j.intfin.2021.101438. URL https://doi.org/10.1016/j.intfin.2021.101438.

Gregory J. Martin and Joshua McCrain. Local news and national politics. *American Political Science Review*, 113(2):372–384, 2019. doi: 10.1017/S0003055418000965.

Maarten Meeuwis, Jonathan A. Parker, Antoinette Schoar, and Duncan Simester. Belief disagreement and portfolio choice. *Journal of Finance*, 77(6):3191–3247, 2022. doi: 10.1111/jofi.13179.

Marina A. Misev and Patrick Balles. Natural disasters, investor attention, and non-fundamental green asset demand. WWZ Working Paper 2024/07, University of Basel, Faculty of Business and Economics, 2024. URL https://wwz.unibas.ch/fileadmin/user_upload/wwz/00_Professuren/Stutzer_Politische_Oekonomie/Publications/2024_07_Green_Investing.pdf.

Morningstar. Global sustainable fund flows: Q4 and full-year 2025 in review. Research report, Morningstar Sustainalytics, February 2026. URL https://www.morningstar.com/content/cs-assets/v3/assets/blt9415ea4cc4157833/blt1d54e64f88b82b3b/Global_ESG_Flows_Q4_2025_Report.pdf.

Stephen Morris and Hyun Song Shin. Social value of public information. *American Economic Review*, 92 (5):1521–1534, 2002. doi: 10.1257/000282802762024610.

Rabih Moussawi, Ke Shen, and Raisa Velthuis. The role of taxes in the rise of ETFs. *Review of Financial Studies*, 38(10):2988–3039, 2025. doi: 10.1093/rfs/hhaf044.

Andrew C. W. Myers. Press coverage and accountability in state legislatures. *American Political Science Review*, 120(1):207–225, 2026. doi: 10.1017/S000305542500022X.

Yihui Pan, Elena S. Pikulina, Stephan Siegel, and Tracy Yue Wang. Political divide and partisan portfolio disagreement. Working paper, March 2025, 2025.

Ľuboš Pástor, Robert F. Stambaugh, and Lucian A. Taylor. Dissecting green returns. *Journal of Financial Economics*, 146(2):403–424, 2022. doi: 10.1016/j.jfineco.2022.07.007.

Pew Research Center. Friends, family and neighbors are americans' most common source of local news. Pew Research Center, Short Reads, 2024a. URL https://www.pewresearch.org/short-reads/2024/09/26/friends-family-and-neighbors-are-americans-most-common-source-of-local-news/.

Pew Research Center. Americans' changing relationship with local news. Pew Research Center, Journalism, 2024b. URL https://www.pewresearch.org/journalism/2024/05/07/americans-changing-relationship-with-local-news/.

Arno Riedl and Paul Smeets. Why do investors hold socially responsible mutual funds? *The Journal of Finance*, 72(6):2505–2550, 2017. doi: 10.1111/jofi.12547.

Marco Sammon and John Shim. Do active funds do better in what they trade? Working paper, November 2023; SSRN 4624934, 2023.

Paul Schmidt-Engelbertz and Kaushik Vasudevan. Speculating on higher-order beliefs. *Review of Financial Studies*, 38(8):2434–2466, 2025. doi: 10.1093/rfs/hhaf019.

Joshua Schwartzstein and Adi Sunderam. Using models to persuade. *American Economic Review*, 111(1): 276–323, 2021. doi: 10.1257/aer.20191074.

James M. Snyder, Jr. and David Strömberg. Press coverage and political accountability. *Journal of Political Economy*, 118(2):355–408, 2010. doi: 10.1086/652903.

Roberto Tubaldi. Mutual funds' fire sales and the real economy: Evidence from hurricanes. Working paper, 2021.

U.S. Census Bureau. Income in the United States: 2025. Current Population Reports P60-289, U.S. Census Bureau, September 2026. URL https://www.census.gov/library/publications/2026/demo/p60-289.html. Median household income in 2025: $87,460, up 2.6 percent from 2024.

Philippe van der Beck. Flow-driven ESG returns. *The Journal of Finance*, 2026. doi: 10.1111/jofi.70083. URL https://doi.org/10.1111/jofi.70083.

Alexander F. Wagner, Richard J. Zeckhauser, and Alexandre Ziegler. Company stock price reactions to the 2016 election shock: Trump, taxes, and trade. *Journal of Financial Economics*, 2018. NBER Working Paper 23152 (February 2017) and HKS RWP17-005; journal volume and pages to VERIFY (MEMORY: 130(2), 428–451).